



Université Gaston Berger de Saint-Louis



UFR SCIENCES APPLIQUEES ET DE TECHNOLOGIE(SAT)
SECTION D'INFORMATIQUE
MASTER SCIENCE ET TECHNOLOGIE
RESEAU, SECURITE ET SYSTEMES DISTRIBUES(R2SD)



PROJET FINAL RESEAUX HAUT DEBIT

RESEAUX HAUT DEBIT



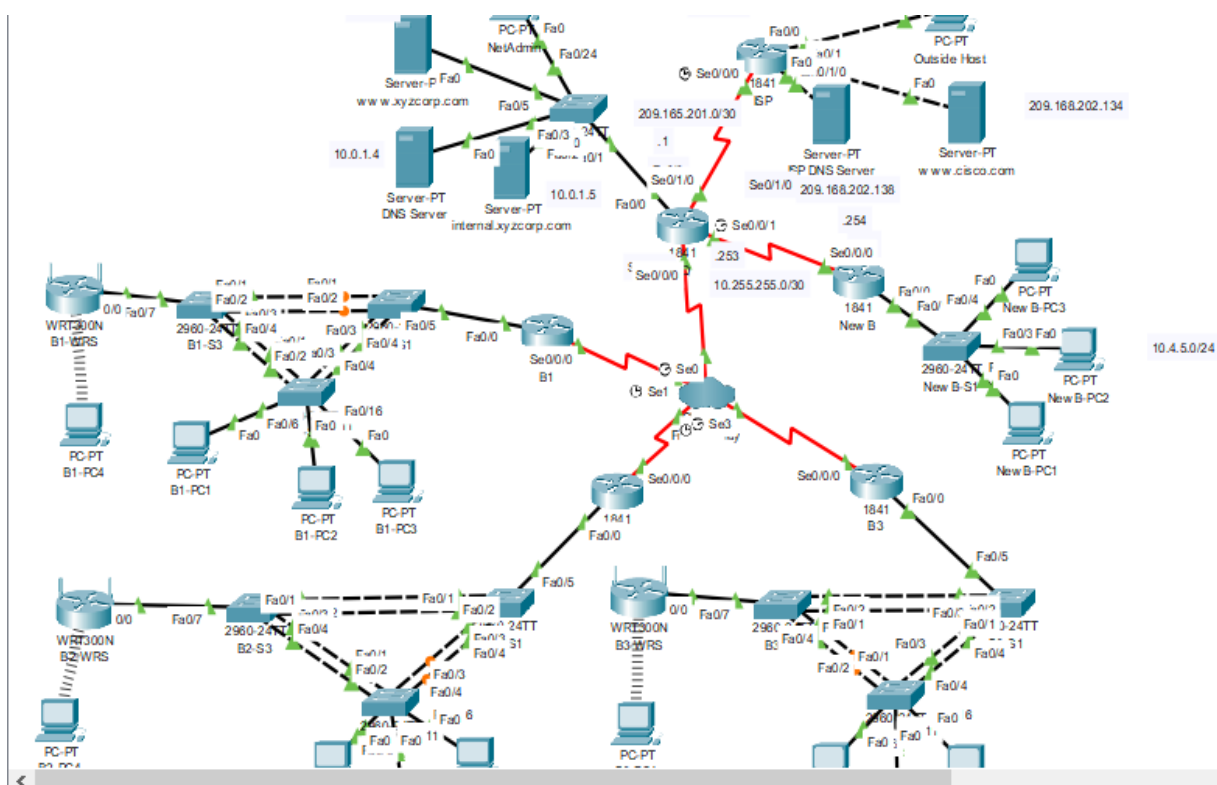
Sous la direction

M. Cheikh Sidy CISSE

Présenté par

Serigne Omar SENE

TOPOLOGIE



Objectifs pédagogiques

Configurer le protocole Frame Relay dans une topologie Hub and Spoke

Configurer l'encapsulation PPP avec authentification CHAP et PAP

Configurer la traduction d'adresses de réseau statique et dynamique

Configurer le routage statique et le routage par défaut

Tâche 1 : configuration du protocole Frame Relay dans une topologie Hub and Spoke

Étape 1 : configuration du cœur de réseau Frame Relay.

Utilisez les tables d'adressage et respectez les consignes suivantes :

HQ est le routeur concentrateur (hub). B1, B2 et B3 sont les rayons (spokes).

HQ utilise une sous-interface point à point pour chaque routeur Branch.

Il est nécessaire d'activer manuellement l'encapsulation IETF sur B3.

Le type LMI doit être configuré manuellement sur la valeur q933a pour HQ, B1 et B2. B3 utilise le protocole ANSI

HQ

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#sh running-config | section fr
Router#sh running-config | section frame
encapsulation frame-relay
frame-relay lmi-type q933a
frame-relay interface-dlci 41
frame-relay interface-dlci 42
frame-relay interface-dlci 43
Router#
```

B1

Physical Config CLI Attributes

IOS Command Line Interface

```
B1#sh running-config | sec
B1#sh running-config | section frame
encapsulation frame-relay
frame-relay interface-dlci 100
frame-relay lmi-type q933a
B1#
```

B2

Physical Config CLI Attributes

IOS Command Line Interface

```
B2#sh running-config | secti
B2#sh running-config | section fr
B2#sh running-config | section fra
B2#sh running-config | section fra
encapsulation frame-relay
frame-relay interface-dlci 200
frame-relay lmi-type q933a
B2#
```

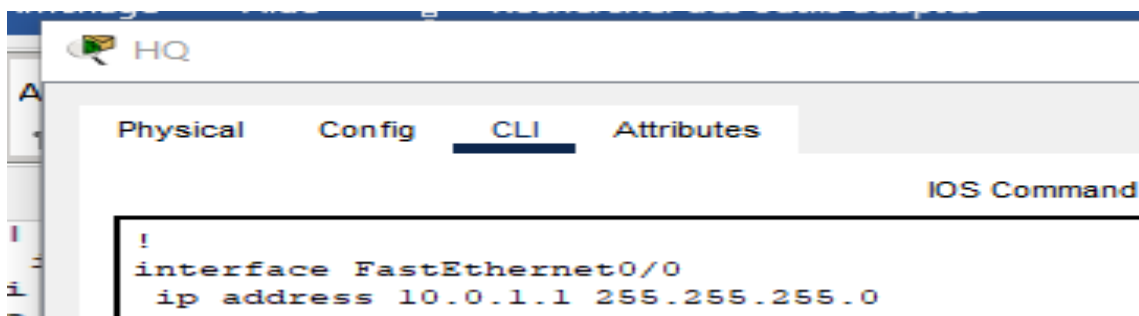
B3

Physical Config CLI Attributes

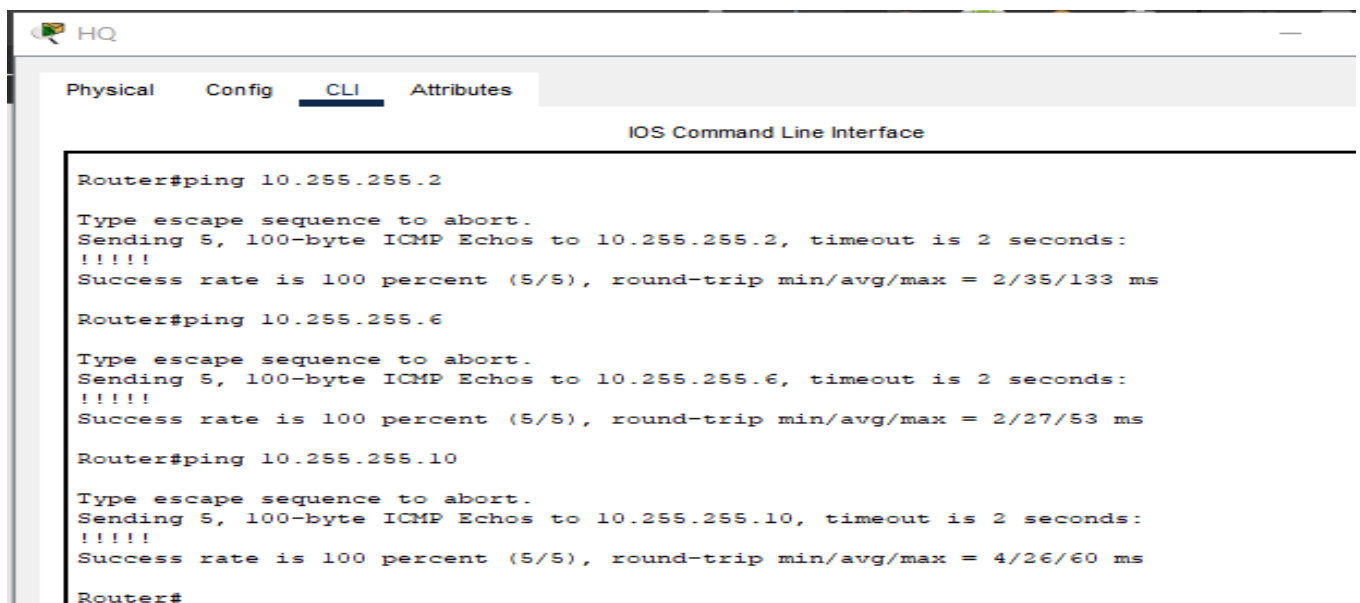
IOS Command Line Interface

```
Router#sh running-config | sect
Router#sh running-config | section frame
encapsulation frame-relay ietf
frame-relay interface-dlci 300
frame-relay lmi-type q933a
Router#
```

Étape 2 : configuration de l'interface de réseau local sur HQ



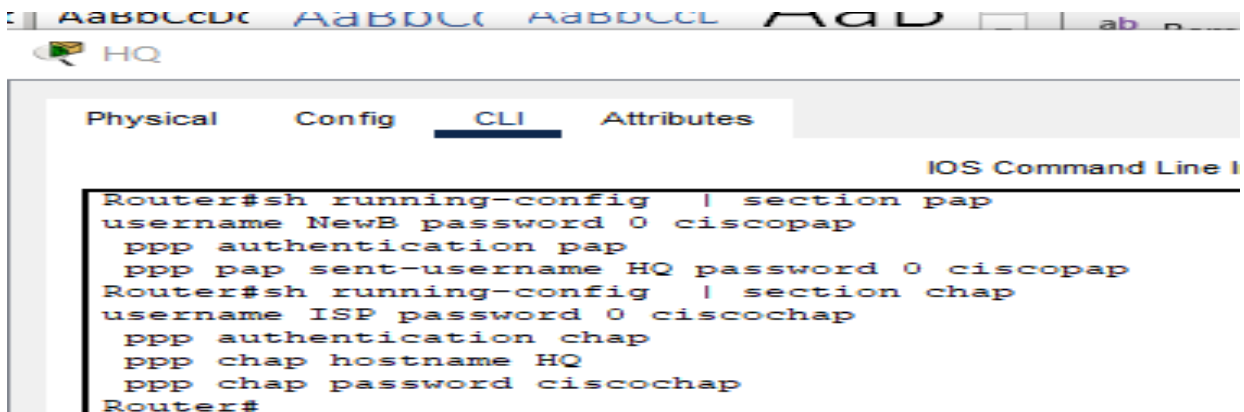
Étape 3 : vérification de l'aboutissement de l'envoi de requêtes ping vers chaque routeur Branch, à partir de HQ



Tâche 2 : configuration de l'encapsulation PPP avec authentification CHAP et PAP

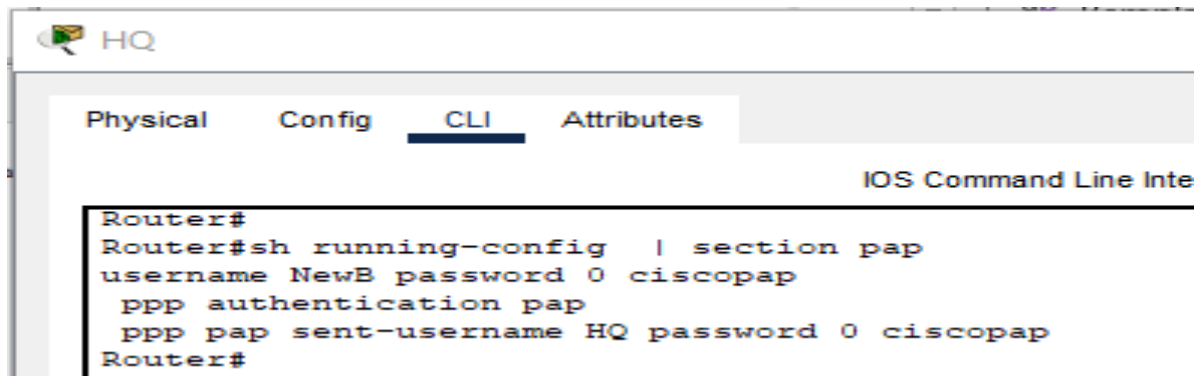
Étape 1 : activation de l'encapsulation PPP avec authentification CHAP sur la liaison WAN reliant HQ et ISP

Le mot de passe CHAP est ciscochap.

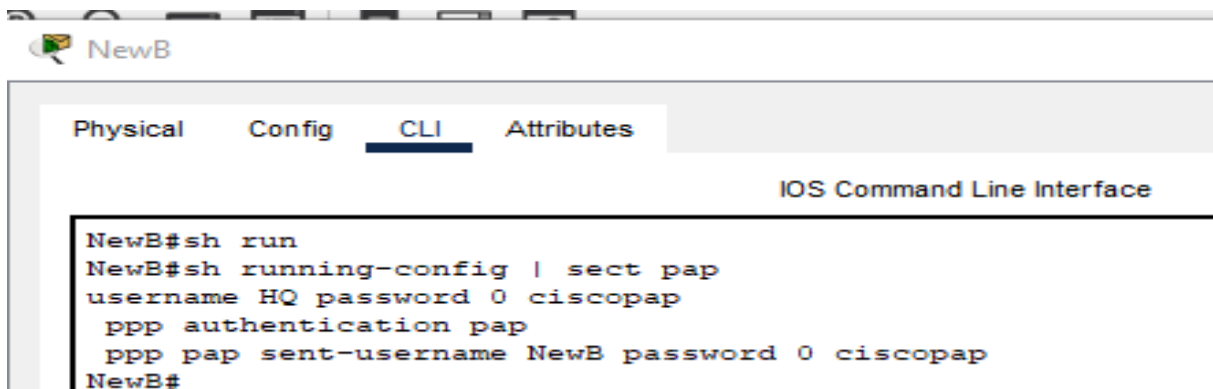


Étape 2 : activation de l'encapsulation PPP avec authentification PAP sur la liaison WAN reliant HQ et NewB

Vous devez connecter un câble aux interfaces adéquates. HQ est l'équipement de communication de données (DCE) de la liaison. Choisissez la fréquence d'horloge. Le mot de passe PAP est ciscopap.

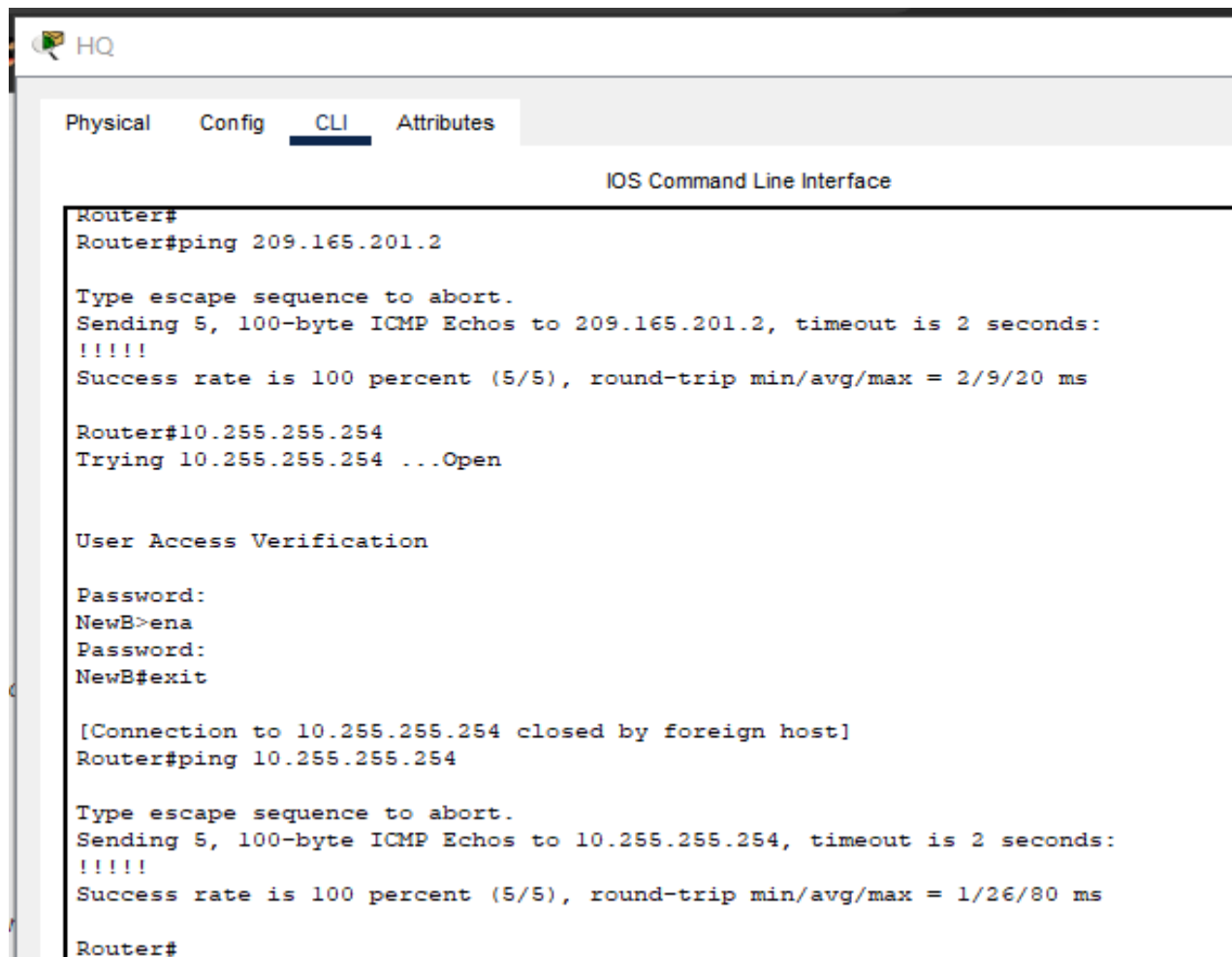


```
Router#
Router#sh running-config | section pap
username NewB password 0 ciscopap
  ppp authentication pap
  ppp pap sent-username HQ password 0 ciscopap
Router#
```



```
NewB#sh run
NewB#sh running-config | sect pap
username HQ password 0 ciscopap
  ppp authentication pap
  ppp pap sent-username NewB password 0 ciscopap
NewB#
```

Étape 3 : vérification de l'aboutissement des envois de requêtes ping vers ISP et NewB à partir de HQ



```

HQ
Physical Config CLI Attributes
IOS Command Line Interface
Router#
Router#ping 209.165.201.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/9/20 ms

Router#10.255.255.254
Trying 10.255.255.254 ...Open

User Access Verification

Password:
NewB>ena
Password:
NewB#exit

[Connection to 10.255.255.254 closed by foreign host]
Router#ping 10.255.255.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.254, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/26/80 ms

Router#

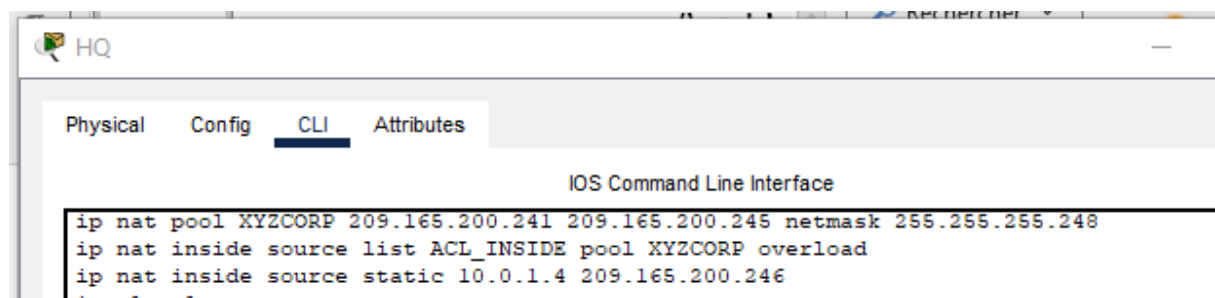
```

Tâche 3 : configuration de la traduction d'adresses de réseau statique et dynamique sur HQ

Étape 1 : configuration de la traduction d'adresses de réseau

Respectez les consignes suivantes :

- Autorisez la traduction de toutes les adresses de l'espace d'adressage 10.0.0.0/8.
- XYZ Corporation détient l'espace d'adressage 209.165.200.240/29. Le pool d'adresses XYZCORP utilise les adresses .241 à .245 avec un masque /29.
- Le site Web www.xyzcorp.com, dont l'adresse est 10.0.1.2, est inscrit auprès du système DNS public à l'adresse IP 209.165.200.246.



```

HQ
Physical Config CLI Attributes
IOS Command Line Interface
ip nat pool XYZCORP 209.165.200.241 209.165.200.245 netmask 255.255.255.248
ip nat inside source list ACL_INSIDE pool XYZCORP overload
ip nat inside source static 10.0.1.4 209.165.200.246
ip nat inside

```

```

!
ip access-list standard ACL_INSIDE
 permit 10.0.0.0 0.255.255.255

```

Étape 2 : vérification du fonctionnement de la traduction d'adresses de réseau par l'envoi de requêtes ping étendues

À partir de HQ, envoyez des requêtes ping vers l'interface série 0/0/0 d'ISP en utilisant comme adresse source l'interface du réseau local HQ. En principe, cet envoi de requêtes ping doit aboutir.

À l'aide de la commande `show ip nat translations`, vérifiez que la traduction d'adresses de réseau a bien traduit l'envoi de requêtes ping.

```

* incomplete command.
Router#sh running-config | section ip nat
ip nat inside
ip nat inside
ip nat inside
ip nat inside
ip nat inside
ip nat outside
ip nat pool XYZCORP 209.165.200.241 209.165.200.245 netmask 255.255.255.248
ip nat inside source list ACL_INSIDE pool XYZCORP overload
ip nat inside source static 10.0.1.4 209.165.200.246
Router#

```

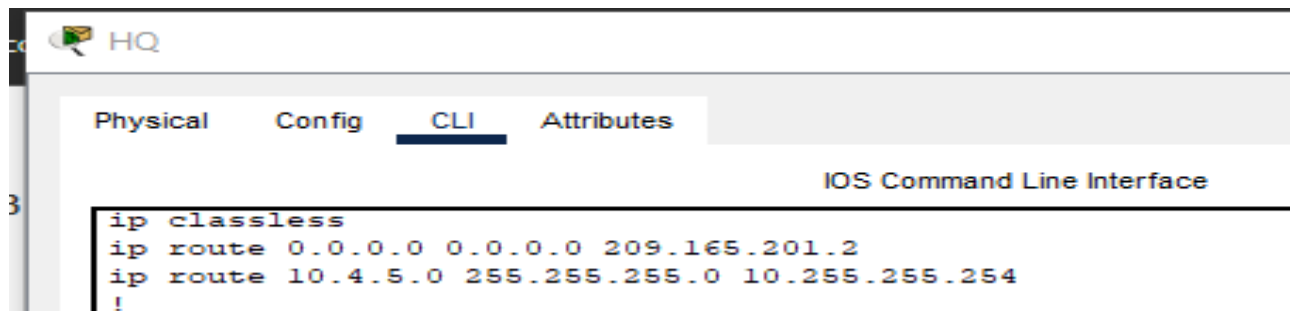
The screenshot displays two windows from a network simulation. On the left, a 'NetAdmin' window shows a Windows Command Prompt where a ping command is executed: `C:\>ping www.cisco.com`. The output shows four successful replies from 209.165.202.134 with varying round trip times (3ms, 5ms, 2ms, 1ms) and a 0% loss. On the right, the 'HQ' window shows the Cisco IOS Command Line Interface. The command `Router#sh ip nat tra` is entered, followed by `Router#sh ip nat translations`, which displays the NAT translation table.

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.165.200.241:4010	0.1.10:40	209.165.202.134:40	209.165.202.134:40
icmp	209.165.200.241:4110	0.1.10:41	209.165.202.134:41	209.165.202.134:41
icmp	209.165.200.241:4210	0.1.10:42	209.165.202.134:42	209.165.202.134:42
icmp	209.165.200.241:4310	0.1.10:43	209.165.202.134:43	209.165.202.134:43
---	209.165.200.246	10.0.1.4	---	---

Tâche 4 : configuration du routage statique et du routage par défaut

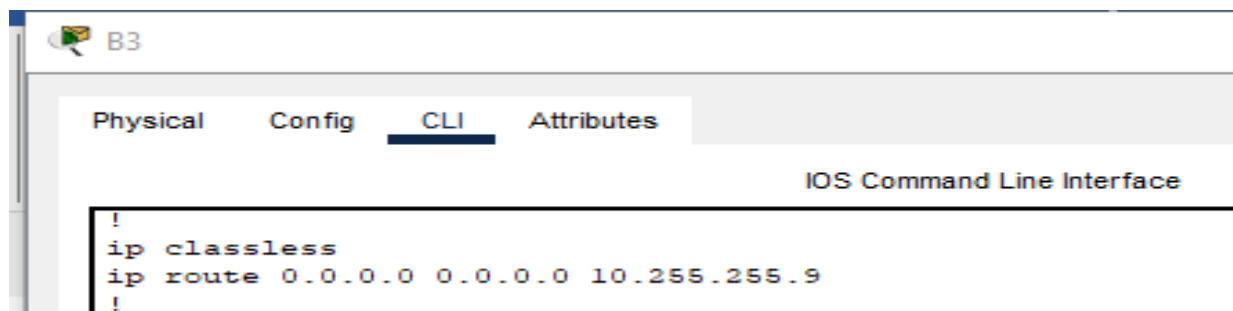
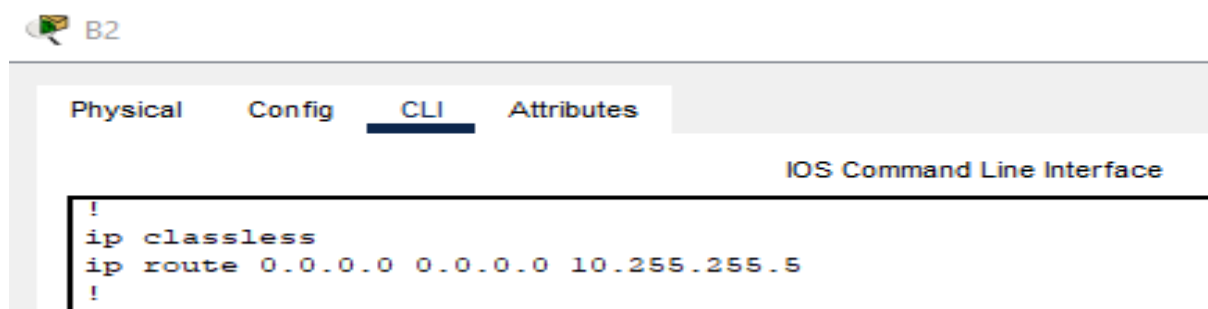
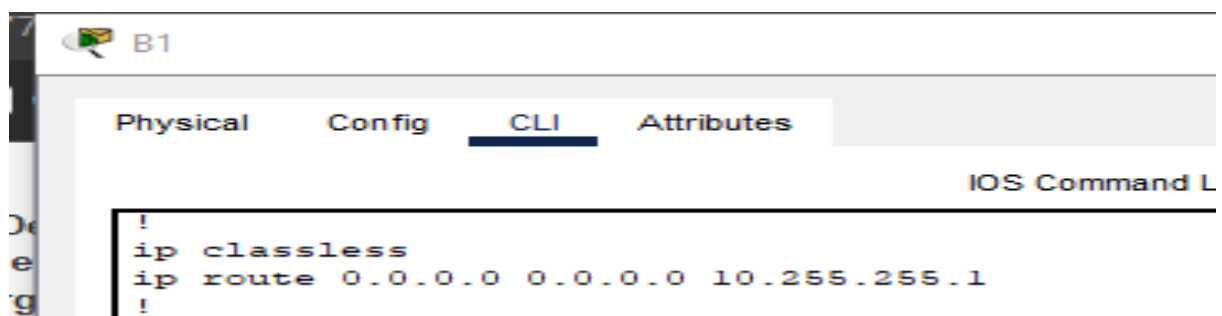
Étape 1 : configuration d'une route par défaut vers ISP et d'une route statique vers le réseau local NewB sur HQ

Utilisez l'argument exit interface.



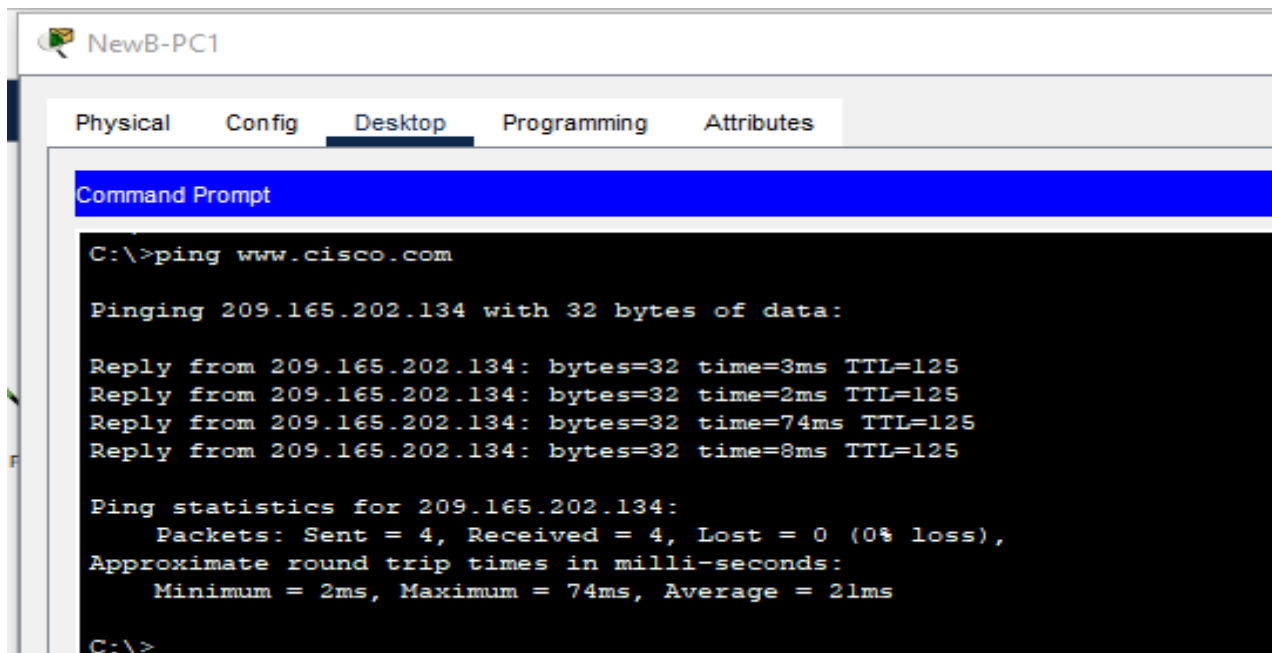
Étape 2 : configuration d'une route par défaut vers HQ sur les routeurs Branch

Utilisez l'adresse IP du tronçon suivant comme argument.



Étape 3 : vérification de la connectivité au-delà d'ISP

Les trois PC NewB et le PC NetAdmin doivent pouvoir envoyer des requêtes ping vers le serveur Web www.cisco.com.



```

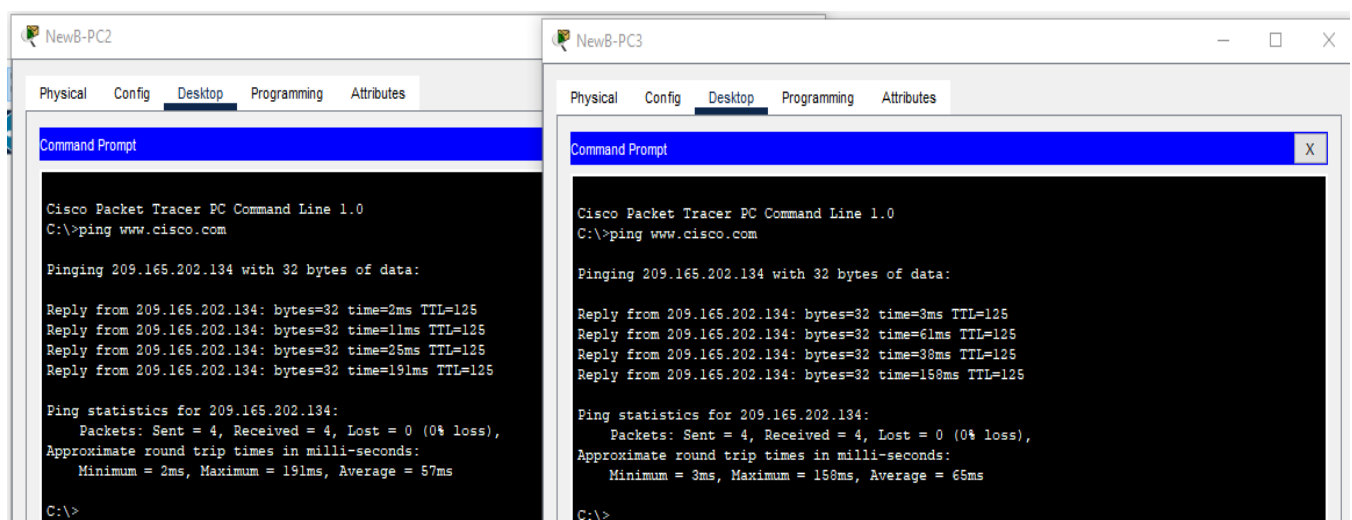
NewB-PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=3ms TTL=125
Reply from 209.165.202.134: bytes=32 time=2ms TTL=125
Reply from 209.165.202.134: bytes=32 time=74ms TTL=125
Reply from 209.165.202.134: bytes=32 time=8ms TTL=125

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 74ms, Average = 21ms

C:\>
  
```



```

NewB-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=2ms TTL=125
Reply from 209.165.202.134: bytes=32 time=11ms TTL=125
Reply from 209.165.202.134: bytes=32 time=25ms TTL=125
Reply from 209.165.202.134: bytes=32 time=191ms TTL=125

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 191ms, Average = 57ms

C:\>

NewB-PC3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=3ms TTL=125
Reply from 209.165.202.134: bytes=32 time=61ms TTL=125
Reply from 209.165.202.134: bytes=32 time=38ms TTL=125
Reply from 209.165.202.134: bytes=32 time=158ms TTL=125

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 158ms, Average = 65ms

C:\>
  
```

Tâche 5 : configuration du routage entre les réseaux locaux virtuels

Étape 1 : configuration du routage entre les réseaux locaux virtuels sur chaque routeur Branch

À l'aide de la table d'adressage des routeurs Branch, configurez et activez le routage entre les réseaux locaux virtuels sur l'interface de réseau local. Le réseau local virtuel VLAN 99 est le VLAN natif.

B1

Physical Config CLI Attributes

IOS Command Line Interface

```
B1#sh running-config | section int
B1#sh running-config | section int
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
interface FastEthernet0/0.10
  encapsulation dot1Q 10
  ip address 10.1.10.1 255.255.255.0
interface FastEthernet0/0.20
  encapsulation dot1Q 20
  ip address 10.1.20.1 255.255.255.0
interface FastEthernet0/0.30
  encapsulation dot1Q 30
  ip address 10.1.30.1 255.255.255.0
interface FastEthernet0/0.88
  encapsulation dot1Q 88
  ip address 10.1.88.1 255.255.255.0
interface FastEthernet0/0.99
  encapsulation dot1Q 99 native
  ip address 10.1.99.1 255.255.255.0
```

B2

Physical Config CLI Attributes

IOS Command Line Interface

```
B2#sh running-config | section int
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
interface FastEthernet0/0.10
  encapsulation dot1Q 10
  ip address 10.2.10.1 255.255.255.0
interface FastEthernet0/0.20
  encapsulation dot1Q 20
  ip address 10.2.20.1 255.255.255.0
interface FastEthernet0/0.30
  encapsulation dot1Q 30
  ip address 10.2.30.1 255.255.255.0
interface FastEthernet0/0.88
  encapsulation dot1Q 88
  ip address 10.2.88.1 255.255.255.0
interface FastEthernet0/0.99
  encapsulation dot1Q 99 native
  ip address 10.2.99.1 255.255.255.0
```

```

Router#sh running-config | section int
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
interface FastEthernet0/0.10
  encapsulation dot1Q 10
  ip address 10.3.10.1 255.255.255.0
interface FastEthernet0/0.20
  encapsulation dot1Q 20
  ip address 10.3.20.1 255.255.255.0
interface FastEthernet0/0.30
  encapsulation dot1Q 30
  ip address 10.3.30.1 255.255.255.0
interface FastEthernet0/0.88
  encapsulation dot1Q 88
  ip address 10.3.88.1 255.255.255.0
interface FastEthernet0/0.99
  encapsulation dot1Q 99 native
  ip address 10.3.99.1 255.255.255.0

```

Étape 2 : vérification des tables de routage

Chaque routeur Branch doit désormais disposer de six réseaux directement connectés et d'une route par défaut statique.

```

B1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.1, 01:01:57, Serial10/0/0
D       10.1.0.0/16 is a summary, 01:01:59, Null0
C       10.1.10.0/24 is directly connected, FastEthernet0/0.10
C       10.1.20.0/24 is directly connected, FastEthernet0/0.20
C       10.1.30.0/24 is directly connected, FastEthernet0/0.30
C       10.1.88.0/24 is directly connected, FastEthernet0/0.88
C       10.1.99.0/24 is directly connected, FastEthernet0/0.99
D       10.2.0.0/16 [90/2684416] via 10.255.255.1, 00:59:03, Serial10/0/0
D       10.3.0.0/16 [90/2684416] via 10.255.255.1, 01:01:54, Serial10/0/0
C       10.255.255.0/30 is directly connected, Serial10/0/0
D       10.255.255.4/30 [90/2681856] via 10.255.255.1, 01:01:57, Serial10/0/0
D       10.255.255.8/30 [90/2681856] via 10.255.255.1, 01:01:57, Serial10/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.1, 01:01:57, Serial10/0/0
S*     0.0.0.0/0 [1/0] via 10.255.255.1
B1#

```

B2

Physical Config CLI Attributes

IOS Command Line Interface

```

B2#sh ip rou
B2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D    10.0.1.0/24 [90/2172416] via 10.255.255.5, 00:57:52, Serial0/0/0
D    10.1.0.0/16 [90/2684416] via 10.255.255.5, 00:57:52, Serial0/0/0
D    10.2.0.0/16 is a summary, 01:00:48, Null0
C    10.2.10.0/24 is directly connected, FastEthernet0/0.10
C    10.2.20.0/24 is directly connected, FastEthernet0/0.20
C    10.2.30.0/24 is directly connected, FastEthernet0/0.30
C    10.2.88.0/24 is directly connected, FastEthernet0/0.88
C    10.2.99.0/24 is directly connected, FastEthernet0/0.99
D    10.3.0.0/16 [90/2684416] via 10.255.255.5, 00:57:52, Serial0/0/0
D    10.255.255.0/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
C    10.255.255.4/30 is directly connected, Serial0/0/0
D    10.255.255.8/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
D    10.255.255.252/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.5
B2#

```

B3

Physical Config CLI Attributes

IOS Command Line Interface

```

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.9 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D    10.0.1.0/24 [90/2172416] via 10.255.255.9, 01:01:33, Serial0/0/0
D    10.1.0.0/16 [90/2684416] via 10.255.255.9, 01:01:33, Serial0/0/0
D    10.2.0.0/16 [90/2684416] via 10.255.255.9, 00:58:39, Serial0/0/0
D    10.3.0.0/16 is a summary, 01:01:35, Null0
C    10.3.10.0/24 is directly connected, FastEthernet0/0.10
C    10.3.20.0/24 is directly connected, FastEthernet0/0.20
C    10.3.30.0/24 is directly connected, FastEthernet0/0.30
C    10.3.88.0/24 is directly connected, FastEthernet0/0.88
C    10.3.99.0/24 is directly connected, FastEthernet0/0.99
D    10.255.255.0/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
D    10.255.255.4/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
C    10.255.255.8/30 is directly connected, Serial0/0/0
D    10.255.255.252/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.9
Router#

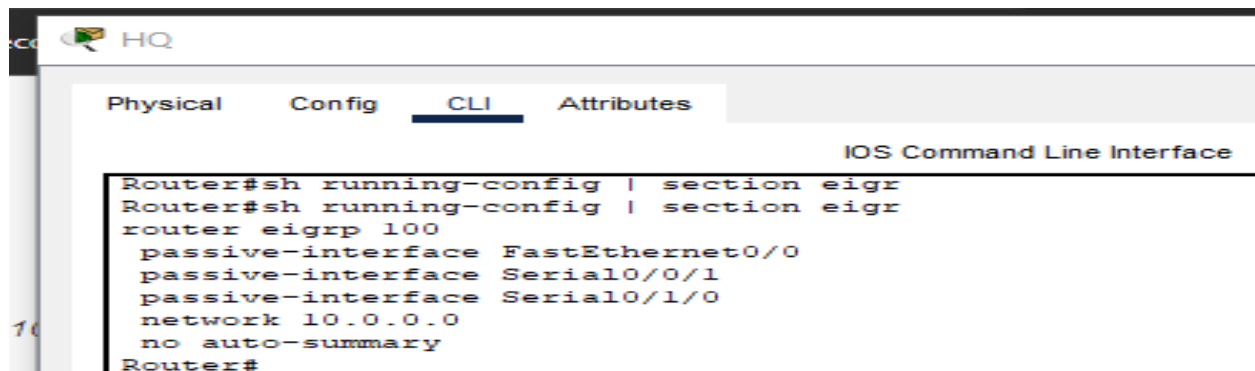
```

Tâche 6 : configuration et optimisation du routage EIGRP

Étape 1 : configuration du protocole EIGRP sur HQ, B1, B2 et B3

- Utilisez AS 100.
- Désactivez les mises à jour EIGRP sur les interfaces adéquates.
- Résumez manuellement le routage EIGRP de manière à ce que chaque routeur Branch annonce uniquement l'espace d'adressage 10.X.0.0/16 à HQ.

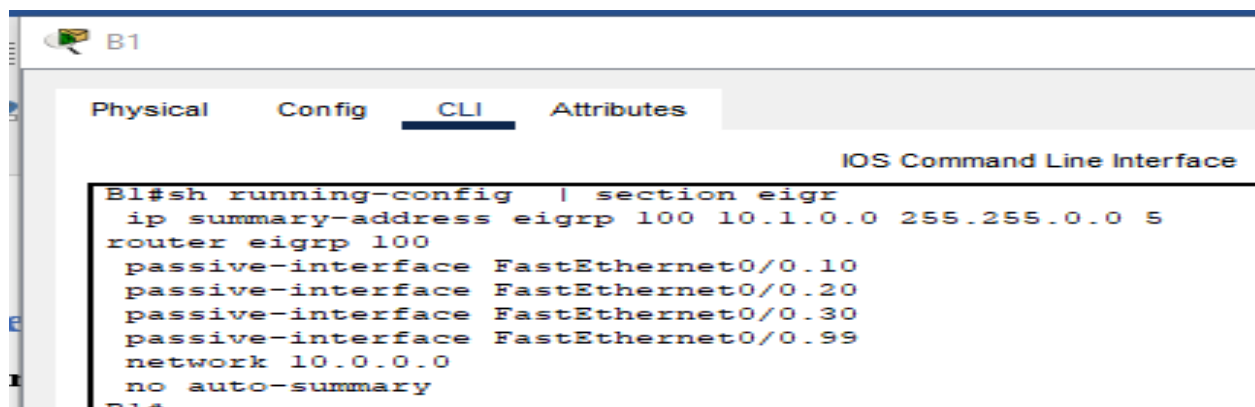
Remarque : Packet Tracer ne simule pas correctement les avantages liés aux résumés du routage EIGRP. Les tables de routage continueront à afficher tous les sous-réseaux, même si vous avez configuré correctement le résumé du routage.



Physical Config **CLI** Attributes

IOS Command Line Interface

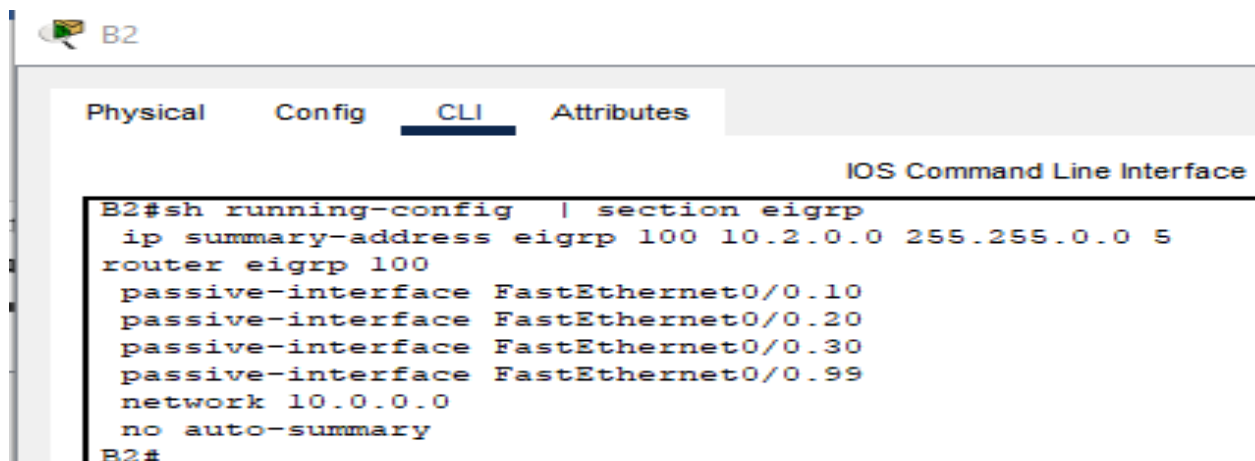
```
Router#sh running-config | section eigr
Router#sh running-config | section eigr
router eigrp 100
  passive-interface FastEthernet0/0
  passive-interface Serial10/0/1
  passive-interface Serial10/1/0
  network 10.0.0.0
  no auto-summary
Router#
```



Physical Config **CLI** Attributes

IOS Command Line Interface

```
B1#sh running-config | section eigr
  ip summary-address eigrp 100 10.1.0.0 255.255.0.0 5
router eigrp 100
  passive-interface FastEthernet0/0.10
  passive-interface FastEthernet0/0.20
  passive-interface FastEthernet0/0.30
  passive-interface FastEthernet0/0.99
  network 10.0.0.0
  no auto-summary
B1#
```



Physical Config **CLI** Attributes

IOS Command Line Interface

```
B2#sh running-config | section eigrp
  ip summary-address eigrp 100 10.2.0.0 255.255.0.0 5
router eigrp 100
  passive-interface FastEthernet0/0.10
  passive-interface FastEthernet0/0.20
  passive-interface FastEthernet0/0.30
  passive-interface FastEthernet0/0.99
  network 10.0.0.0
  no auto-summary
B2#
```



```

Router#sh running-config | section eigrp
ip summary-address eigrp 100 10.3.0.0 255.255.0.0 5
router eigrp 100
passive-interface FastEthernet0/0.10
passive-interface FastEthernet0/0.20
passive-interface FastEthernet0/0.30
passive-interface FastEthernet0/0.99
network 10.0.0.0
no auto-summary
Router#

```

Étape 2 : vérification des tables de routage et de la connectivité

Les routeurs HQ et Branch doivent désormais disposer de tables de routage complètes.

Le PC NetAdmin doit désormais être capable d'envoyer des paquets ping vers chacune des sous-interfaces VLAN des routeurs Branch.

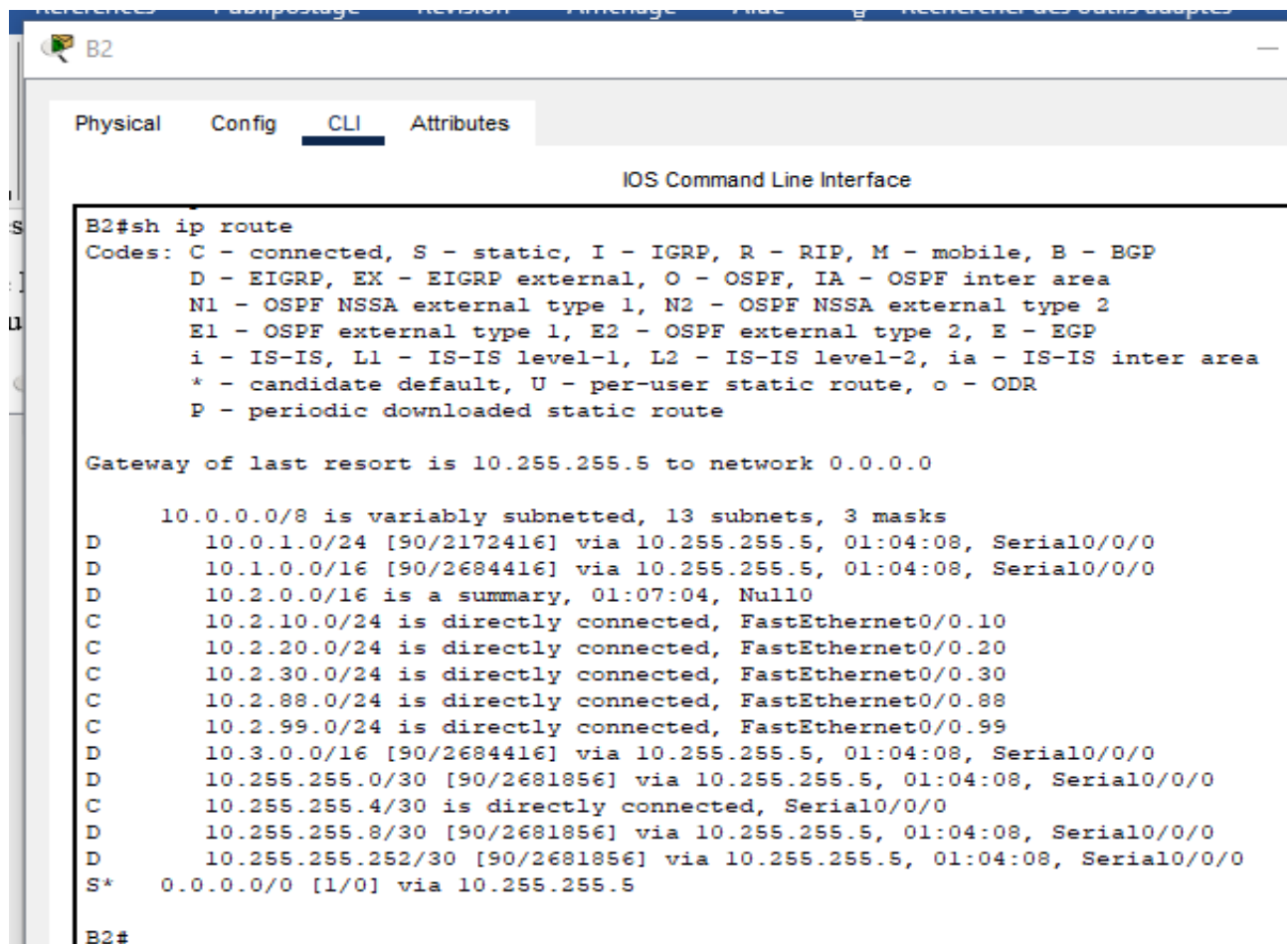
```

B1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.1, 01:06:38, Serial0/0/0
D       10.1.0.0/16 is a summary, 01:06:40, Null0
C       10.1.10.0/24 is directly connected, FastEthernet0/0.10
C       10.1.20.0/24 is directly connected, FastEthernet0/0.20
C       10.1.30.0/24 is directly connected, FastEthernet0/0.30
C       10.1.88.0/24 is directly connected, FastEthernet0/0.88
C       10.1.99.0/24 is directly connected, FastEthernet0/0.99
D       10.2.0.0/16 [90/2684416] via 10.255.255.1, 01:03:44, Serial0/0/0
D       10.3.0.0/16 [90/2684416] via 10.255.255.1, 01:06:35, Serial0/0/0
C       10.255.255.0/30 is directly connected, Serial0/0/0
D       10.255.255.4/30 [90/2681856] via 10.255.255.1, 01:06:38, Serial0/0/0
D       10.255.255.8/30 [90/2681856] via 10.255.255.1, 01:06:38, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.1, 01:06:38, Serial0/0/0
S*     0.0.0.0/0 [1/0] via 10.255.255.1
B1#

```



B2

Physical Config **CLI** Attributes

IOS Command Line Interface

```

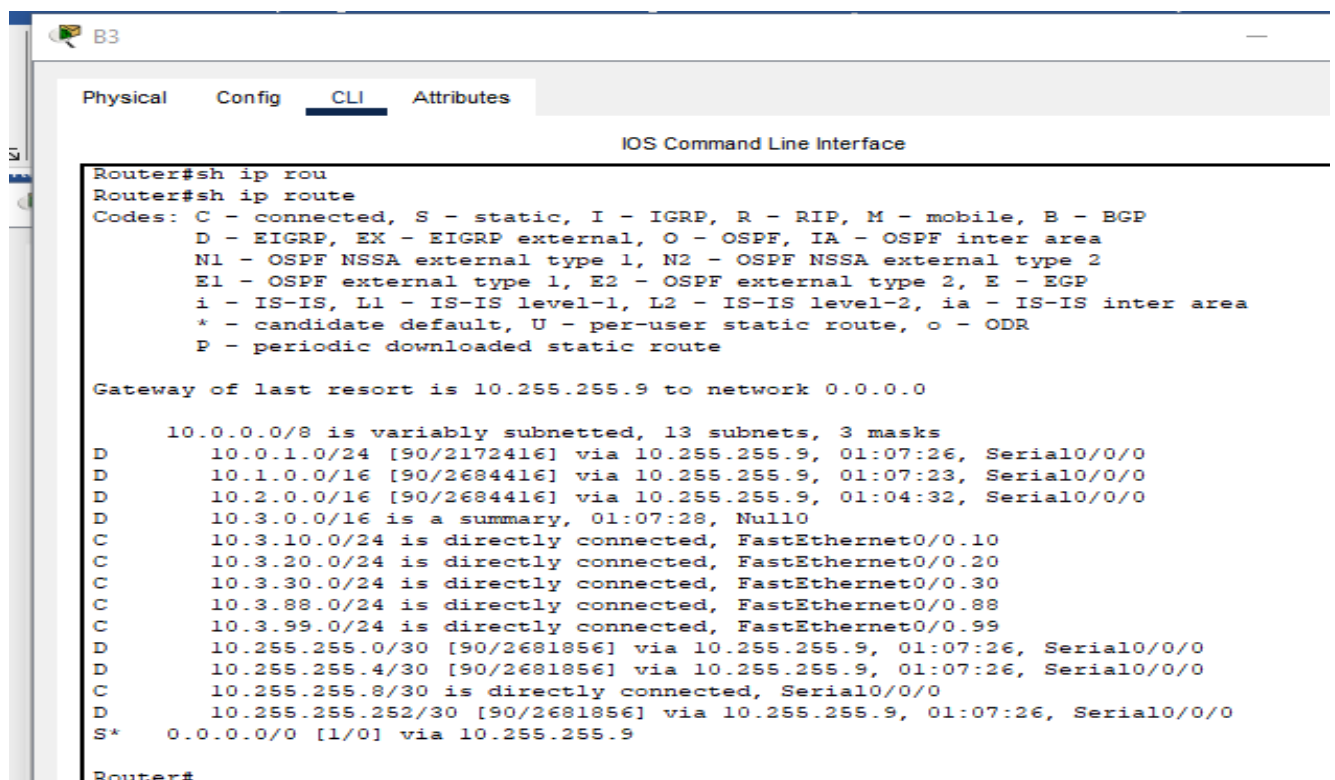
B2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.2.0.0/16 is a summary, 01:07:04, Null0
C       10.2.10.0/24 is directly connected, FastEthernet0/0.10
C       10.2.20.0/24 is directly connected, FastEthernet0/0.20
C       10.2.30.0/24 is directly connected, FastEthernet0/0.30
C       10.2.88.0/24 is directly connected, FastEthernet0/0.88
C       10.2.99.0/24 is directly connected, FastEthernet0/0.99
D       10.3.0.0/16 [90/2684416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.255.255.0/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
C       10.255.255.4/30 is directly connected, Serial0/0/0
D       10.255.255.8/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
S*    0.0.0.0/0 [1/0] via 10.255.255.5

B2#

```



B3

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Router#sh ip rou
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.9 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.9, 01:07:26, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.9, 01:07:23, Serial0/0/0
D       10.2.0.0/16 [90/2684416] via 10.255.255.9, 01:04:32, Serial0/0/0
D       10.3.0.0/16 is a summary, 01:07:28, Null0
C       10.3.10.0/24 is directly connected, FastEthernet0/0.10
C       10.3.20.0/24 is directly connected, FastEthernet0/0.20
C       10.3.30.0/24 is directly connected, FastEthernet0/0.30
C       10.3.88.0/24 is directly connected, FastEthernet0/0.88
C       10.3.99.0/24 is directly connected, FastEthernet0/0.99
D       10.255.255.0/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
D       10.255.255.4/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
C       10.255.255.8/30 is directly connected, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
S*    0.0.0.0/0 [1/0] via 10.255.255.9

Router#

```

Test de connectivité

```

C:\>ping 10.3.10.1

Pinging 10.3.10.1 with 32 bytes of data:

Reply from 10.3.10.1: bytes=32 time=2ms TTL=254
Reply from 10.3.10.1: bytes=32 time=20ms TTL=254
Reply from 10.3.10.1: bytes=32 time=2ms TTL=254
Reply from 10.3.10.1: bytes=32 time=2ms TTL=254

Ping statistics for 10.3.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 20ms, Average = 6ms

C:\>ping 10.3.20.1

Pinging 10.3.20.1 with 32 bytes of data:

Reply from 10.3.20.1: bytes=32 time=96ms TTL=254
Reply from 10.3.20.1: bytes=32 time=3ms TTL=254
Reply from 10.3.20.1: bytes=32 time=51ms TTL=254
Reply from 10.3.20.1: bytes=32 time=3ms TTL=254

Ping statistics for 10.3.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 96ms, Average = 38ms

C:\>ping 10.3.30.1

Pinging 10.3.30.1 with 32 bytes of data:

Reply from 10.3.30.1: bytes=32 time=2ms TTL=254
Reply from 10.3.30.1: bytes=32 time=2ms TTL=254
Reply from 10.3.30.1: bytes=32 time=2ms TTL=254
Reply from 10.3.30.1: bytes=32 time=50ms TTL=254

Ping statistics for 10.3.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 50ms, Average = 14ms

C:\>ping 10.2.30.1
  
```

Tâche 7 : configuration du protocole VTP, de l'agrégation, de l'interface VLAN et des réseaux locaux virtuels

Les conditions suivantes s'appliquent aux trois branches. Configurez un jeu de trois commutateurs. Utilisez ensuite les scripts de ces commutateurs sur les deux autres jeux de commutateurs.

Étape 1 : configuration du protocole VTP sur les commutateurs Branch

BX-S1 est le serveur VTP. BX-S2 et BX-S3 sont des clients VTP.

Le nom de domaine est XYZCORP.

Le mot de passe est xyzvtp.

B1-S1

Physical Config CLI Attributes

IOS Command Line Interface

```

Switch#sh vtp status
VTP Version                : 1
Configuration Revision      : 22
Maximum VLANs supported locally : 255
Number of existing VLANs    : 10
VTP Operating Mode          : Server
VTP Domain Name             : XYZCORP
VTP Pruning Mode            : Disabled
VTP V2 Mode                 : Disabled
VTP Traps Generation        : Disabled
MD5 digest                  : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Local updater ID is 10.1.99.21 on interface Vl99 (lowest numbered VLAN interface found)
Switch#
Switch#

```

B1-S2

Physical Config CLI Attributes

IOS Command Line Interface

```

Switch#sh vtp status
VTP Version                : 1
Configuration Revision      : 22
Maximum VLANs supported locally : 255
Number of existing VLANs    : 10
VTP Operating Mode          : Client
VTP Domain Name             : XYZCORP
VTP Pruning Mode            : Disabled
VTP V2 Mode                 : Disabled
VTP Traps Generation        : Disabled
MD5 digest                  : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Switch#

```

B1-S3

Physical Config CLI Attributes

IOS Command Line Interface

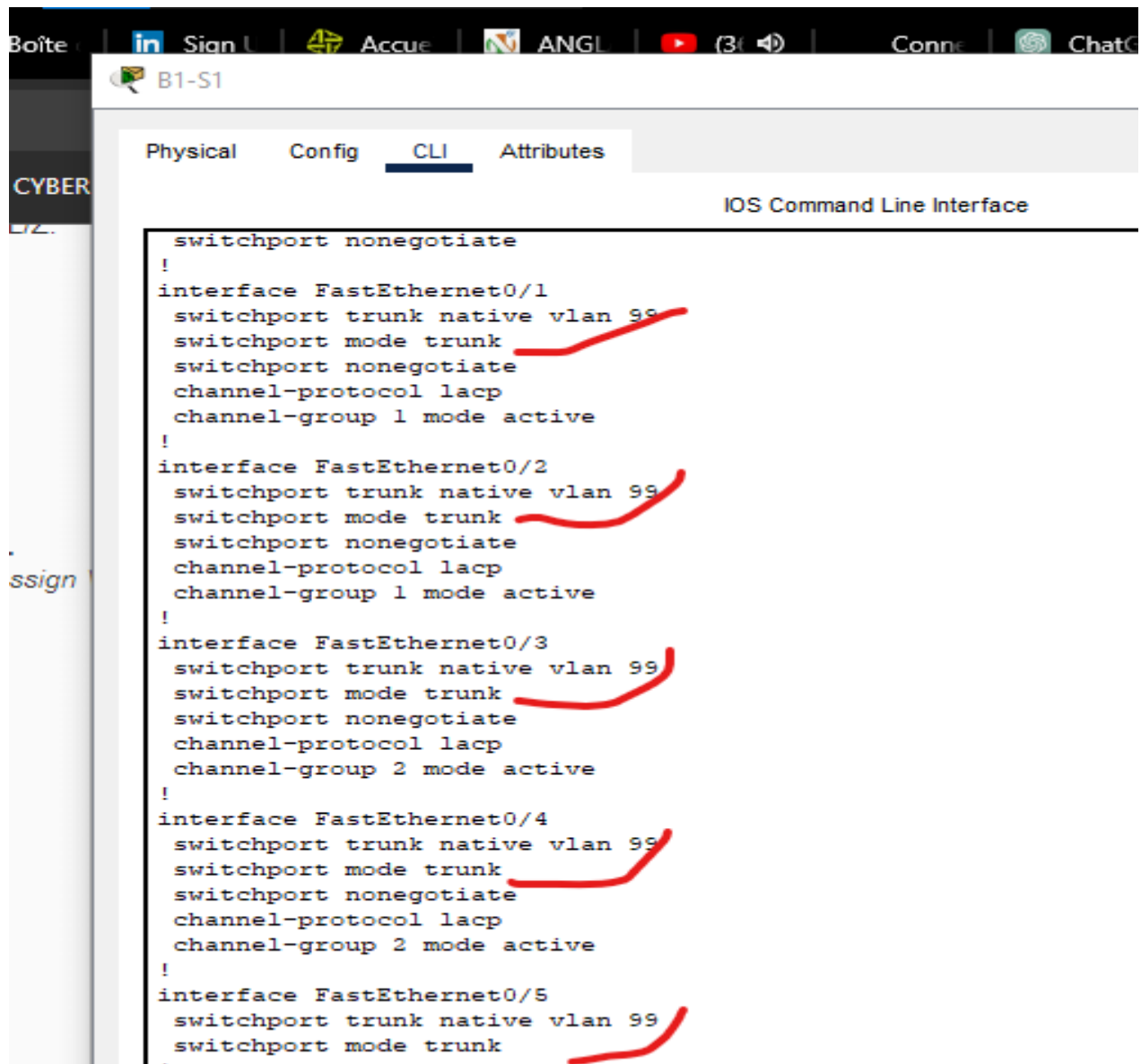
```

Switch#sh vtp status
VTP Version                : 1
Configuration Revision      : 22
Maximum VLANs supported locally : 255
Number of existing VLANs    : 10
VTP Operating Mode          : Client
VTP Domain Name             : XYZCORP
VTP Pruning Mode            : Disabled
VTP V2 Mode                 : Disabled
VTP Traps Generation        : Disabled
MD5 digest                  : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Switch#

```

Étape 2 : configuration de l'agrégation sur BX-S1, BX-S2 et BX-S3

Configurez les interfaces adéquates en mode d'agrégation et désignez le réseau local virtuel VLAN 99 en tant que VLAN natif



```
Boîte | Sign | Accue | ANGL | (3) | Conn | ChatC
B1-S1
Physical Config CLI Attributes
CYBER
IOS Command Line Interface

switchport nonegotiate
!
interface FastEthernet0/1
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode active
!
interface FastEthernet0/2
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode active
!
interface FastEthernet0/3
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 2 mode active
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 2 mode active
!
interface FastEthernet0/5
  switchport trunk native vlan 99
  switchport mode trunk
```

B1-S2

Physical Config CLI Attributes

IOS Command Line Interface

```

switchport nonegotiate
!
interface FastEthernet0/1
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 2 mode active
!
interface FastEthernet0/2
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 2 mode active
!
interface FastEthernet0/3
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode passive
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode passive

```

B1-S3

Physical Config CLI Attributes

IOS Command Line Interface

```

switchport nonegotiate
!
interface FastEthernet0/1
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode passive
!
interface FastEthernet0/2
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 1 mode passive
!
interface FastEthernet0/3
  switchport trunk native vlan 99
  switchport mode trunk
  switchport nonegotiate
  channel-protocol lacp
  channel-group 2 mode passive
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport mode trunk

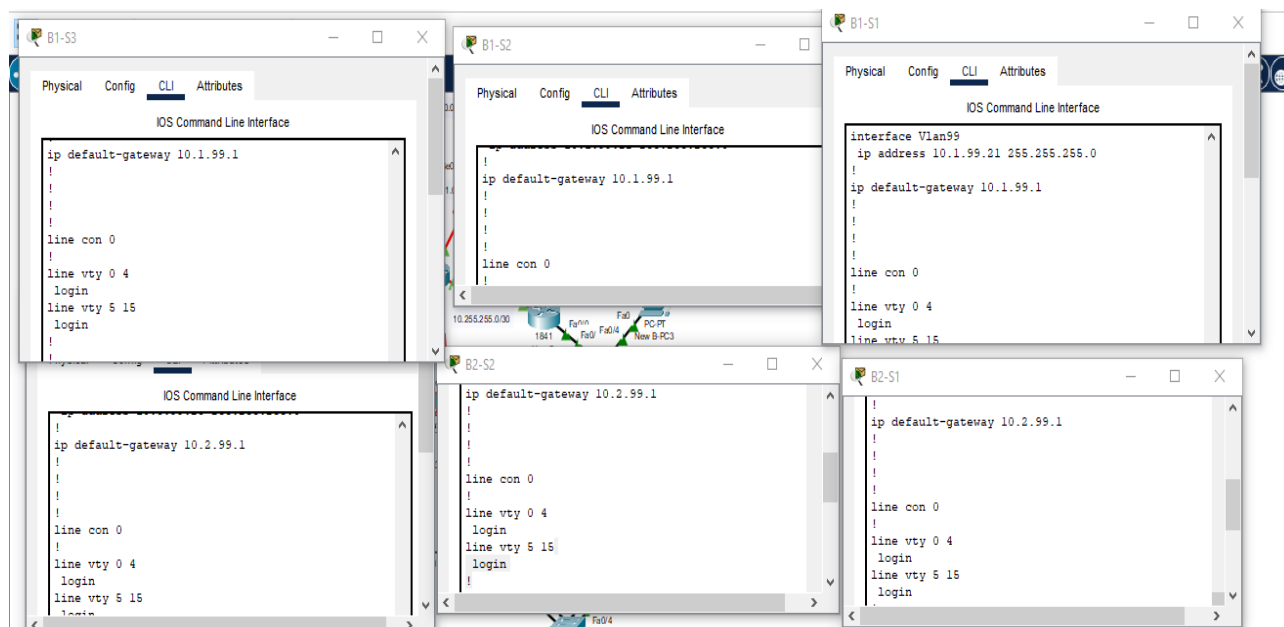
```

Faire la même chose pour :

B2-S1, B2-S2, B2-S3

B3-S1, B3-S2, B3-S3

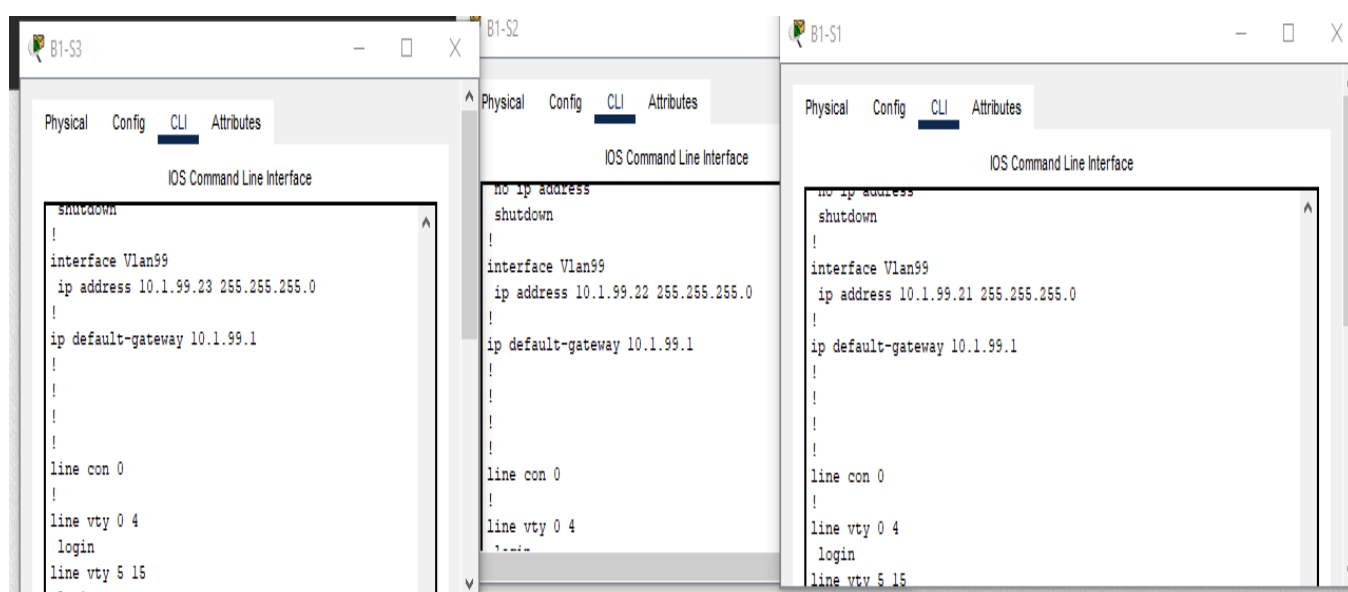
Étape 3 : configuration de l'interface VLAN et de la passerelle par défaut sur BX-S1, BX-S2 et BX-S3



B3-S1 :10.3.99.1

B3-S2 :10.3.99.1

B3-S2 :10.3.99.1



Étape 4 : création des réseaux locaux virtuels (VLAN) sur BX-S1

B1-S1

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch#sh vl
Switch#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Admin	active	
20	Sales	active	
30	Production	active	
88	Wireless	active	
99	Mgmt&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	en	100001	1500	-	-	-	-	-	0	0
10	en	100010	1500	-	-	-	-	-	0	0
20	en	100020	1500	-	-	-	-	-	0	0
30	en	100030	1500	-	-	-	-	-	0	0
88	en	100088	1500	-	-	-	-	-	0	0
99	en	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

B2-S1

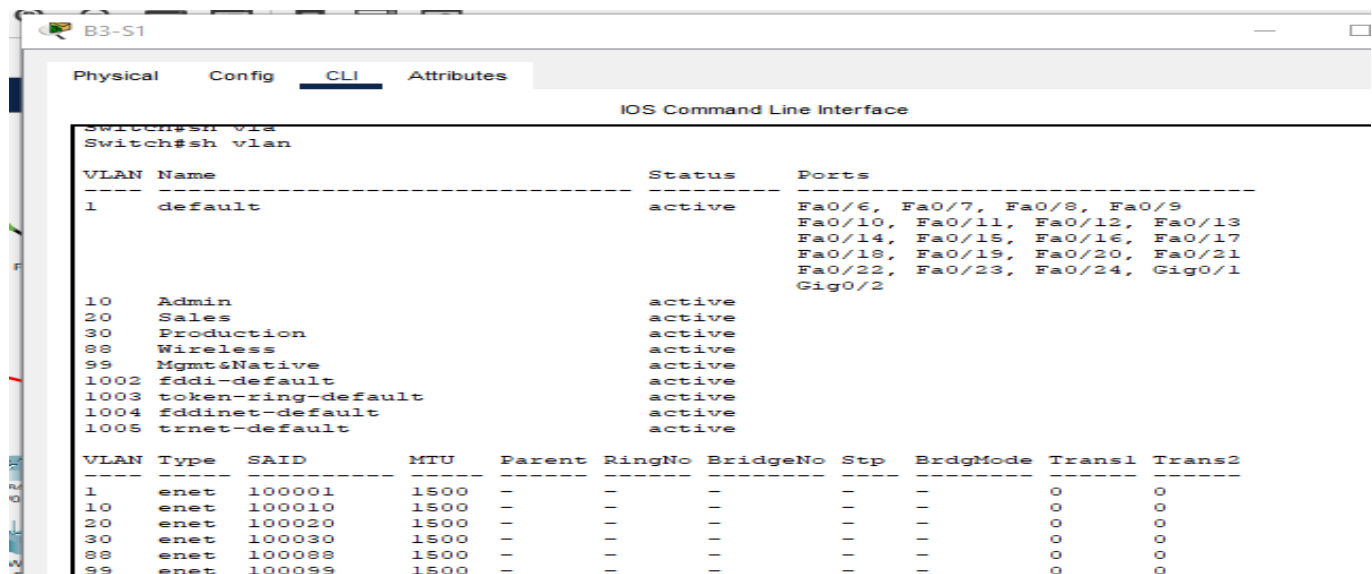
Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Admin	active	
20	Sales	active	
30	Production	active	
88	Wireless	active	
99	Mgmt&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
88	enet	100088	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0



B3-S1

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch#sh vtp
Switch#sh vlan
```

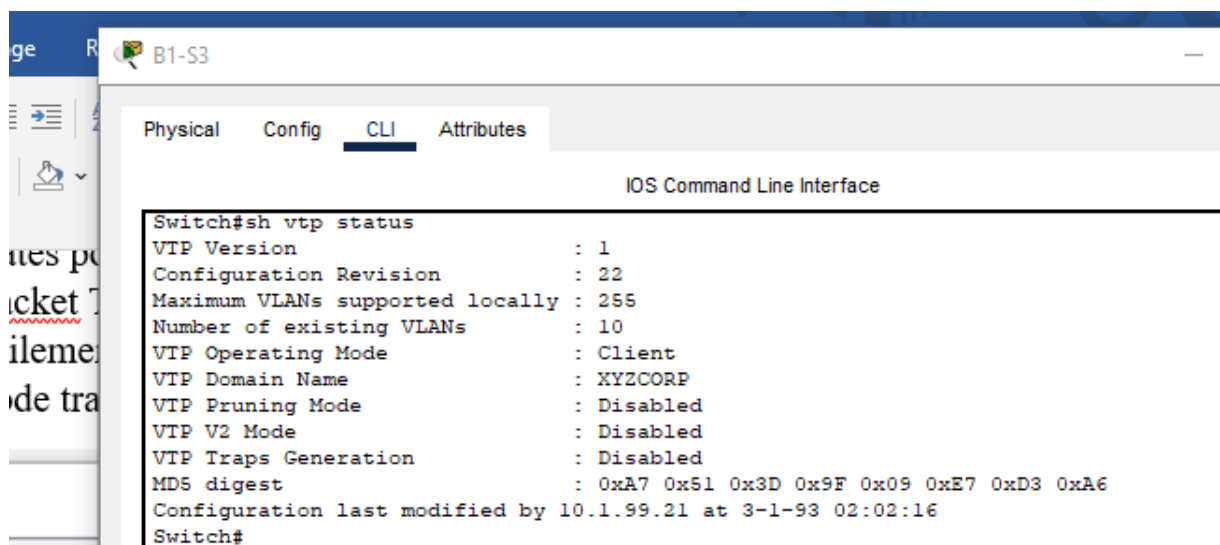
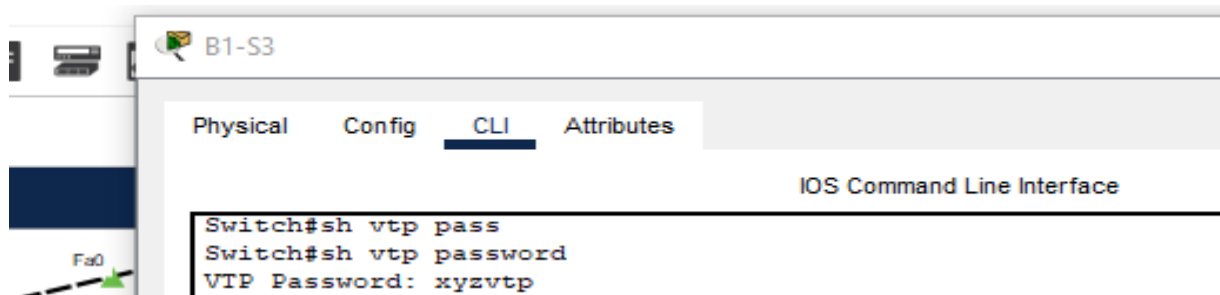
VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Admin	active	
20	Sales	active	
30	Production	active	
88	Wireless	active	
99	Mgmt&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

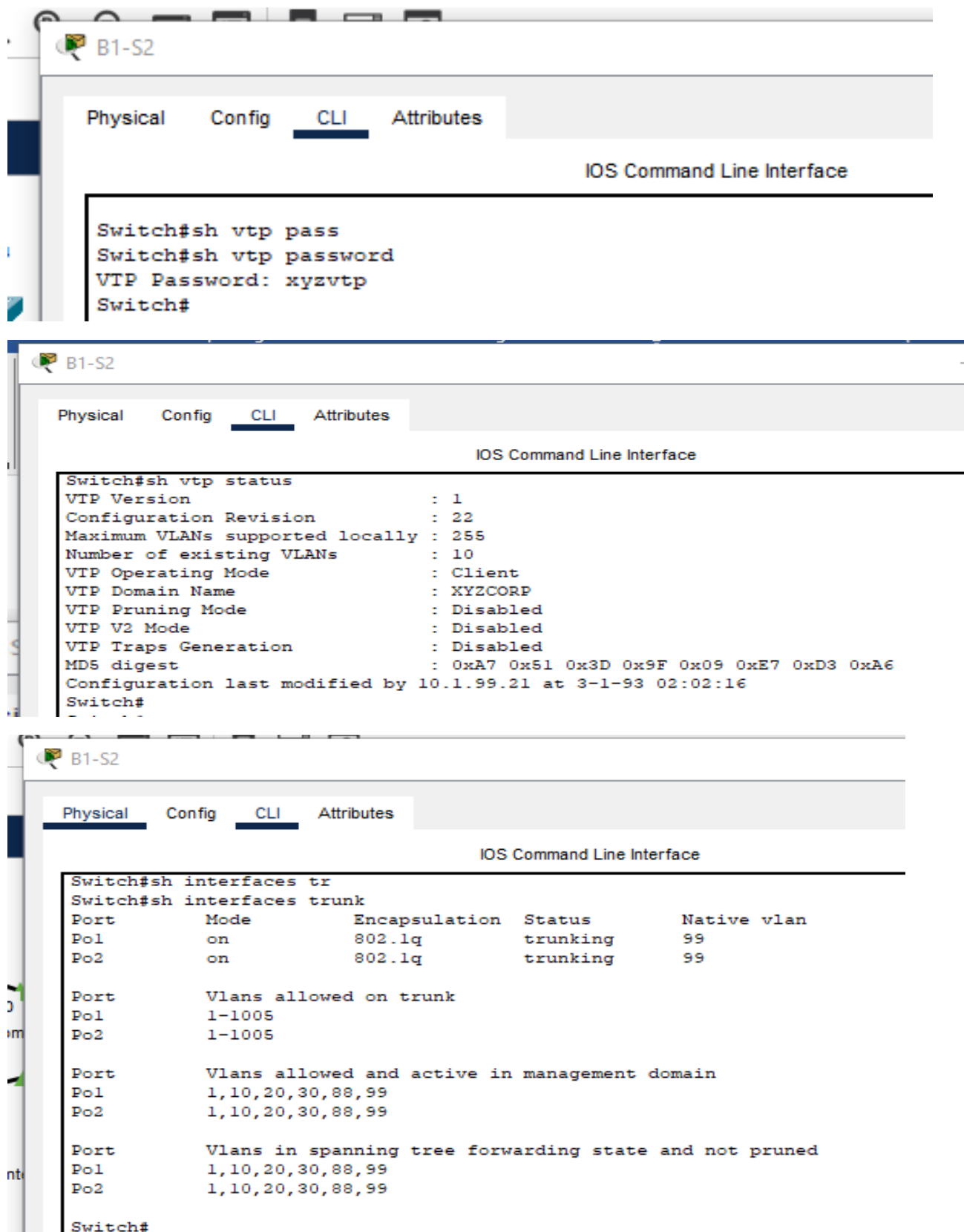
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	1000001	1500	-	-	-	-	-	0	0
10	enet	1000010	1500	-	-	-	-	-	0	0
20	enet	1000020	1500	-	-	-	-	-	0	0
30	enet	1000030	1500	-	-	-	-	-	0	0
88	enet	1000088	1500	-	-	-	-	-	0	0
99	enet	1000099	1500	-	-	-	-	-	0	0

Sur BX-S1 uniquement, créez et nommez les réseaux locaux virtuels (VLAN) répertoriés dans le tableau Configuration VLAN et mappages de ports. VTP annonce les nouveaux réseaux locaux virtuels à BX-S1 et BX-S2.

Étape 5 : vérification de l'envoi des réseaux locaux virtuels à BX-S2 et BX-S3

Utilisez les commandes adéquates pour vérifier que S2 et S3 disposent désormais des réseaux locaux virtuels créés sur S1. Packet Tracer peut mettre plusieurs minutes à simuler les annonces VTP. Pour forcer facilement l'envoi des annonces VTP, vous pouvez basculer un des commutateurs client du mode transparent au mode client et inversement.





Tâche 8 : affectation des réseaux locaux virtuels et configuration de la sécurité des ports

Étape 1 : affectation des réseaux locaux virtuels aux ports d'accès

Utilisez le tableau Configuration VLAN et mappages de ports pour réaliser les opérations suivantes :

- Configurer les ports d'accès
- Affecter des réseaux locaux virtuels aux ports d'accès

B1-S2

Physical Config **CLI** Attributes

IOS Command Line Interface

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Admin	active	Fa0/6
20	Sales	active	Fa0/11
30	Production	active	Fa0/16
88	Wireless	active	
99	Mgmt&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
88	enet	100088	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
88	enet	100088	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

B2-S2

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch#sh vl
Switch#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Admin	active	Fa0/6
20	Sales	active	Fa0/11
30	Production	active	Fa0/16
88	Wireless	active	
99	Mgmt&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
88	enet	100088	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
88	enet	100088	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

```

B3-S2
Physical Config CLI Attributes
IOS Command Line Interface

Switch#sh vlan

VLAN Name                Status Ports
-----
1    default              active Fa0/5, Fa0/7, Fa0/8, Fa0/9
    Fa0/10, Fa0/12, Fa0/13, Fa0/14
    Fa0/15, Fa0/17, Fa0/18, Fa0/19
    Fa0/20, Fa0/21, Fa0/22, Fa0/23
    Fa0/24, Gig0/1, Gig0/2
10   Admin                active Fa0/6
20   Sales                active Fa0/11
30   Production           active Fa0/16
88   Wireless             active
99   Mgmt&Native          active
1002 fddi-default         active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001   1500  -      -      -      -   -        0      0
10   enet   100010   1500  -      -      -      -   -        0      0
20   enet   100020   1500  -      -      -      -   -        0      0
30   enet   100030   1500  -      -      -      -   -        0      0
88   enet   100088   1500  -      -      -      -   -        0      0
99   enet   100099   1500  -      -      -      -   -        0      0
1002 fddi   101002   1500  -      -      -      -   -        0      0
1003 tr    101003   1500  -      -      -      -   -        0      0
1004 fdnet 101004   1500  -      -      -      -   ieee     0      0
1005 trnet 101005   1500  -      -      -      -   ibm      0      0

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----

```

```

B1-S3
Physical Config CLI Attributes
IOS Command Line Interface

Switch>sh vlan

VLAN Name                Status Ports
-----
1    default              active Fa0/5, Fa0/6, Fa0/8, Fa0/9
    Fa0/10, Fa0/11, Fa0/12, Fa0/13
    Fa0/14, Fa0/15, Fa0/16, Fa0/17
    Fa0/18, Fa0/19, Fa0/20, Fa0/21
    Fa0/22, Fa0/23, Fa0/24, Gig0/1
    Gig0/2
10   Admin                active
20   Sales                active
30   Production           active
88   Wireless             active Fa0/7
99   Mgmt&Native          active
1002 fddi-default         active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001   1500  -      -      -      -   -        0      0
10   enet   100010   1500  -      -      -      -   -        0      0
20   enet   100020   1500  -      -      -      -   -        0      0
30   enet   100030   1500  -      -      -      -   -        0      0
88   enet   100088   1500  -      -      -      -   -        0      0
99   enet   100099   1500  -      -      -      -   -        0      0
1002 fddi   101002   1500  -      -      -      -   -        0      0

```

```

B2-S3
Physical Config CLI Attributes
IOS Command Line Interface

Switch>sh vlan

VLAN Name                Status Ports
-----
1    default              active Fa0/5, Fa0/6, Fa0/8, Fa0/9
    Fa0/10, Fa0/11, Fa0/12, Fa0/13
    Fa0/14, Fa0/15, Fa0/16, Fa0/17
    Fa0/18, Fa0/19, Fa0/20, Fa0/21
    Fa0/22, Fa0/23, Fa0/24, Gig0/1
    Gig0/2
10   Admin                active
20   Sales                active
30   Production           active
88   Wireless             active Fa0/7
99   Mgmt&Native          active
1002 fddi-default         active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001   1500  -      -      -      -   -        0      0
10   enet   100010   1500  -      -      -      -   -        0      0
20   enet   100020   1500  -      -      -      -   -        0      0
30   enet   100030   1500  -      -      -      -   -        0      0
88   enet   100088   1500  -      -      -      -   -        0      0
99   enet   100099   1500  -      -      -      -   -        0      0
1002 fddi   101002   1500  -      -      -      -   -        0      0
1003 tr    101003   1500  -      -      -      -   -        0      0
1004 fdnet 101004   1500  -      -      -      -   ieee     0      0
1005 trnet 101005   1500  -      -      -      -   ibm      0      0

```

The image displays three screenshots of network switch CLI interfaces, labeled B3-S3, B1-S2, and B1-S1.

B3-S3 Screenshot: Shows the output of the command `Switch#sh vlan`. It lists VLANs 1 through 1005 with their names, status, and associated ports. A red line highlights the ports for VLAN 1005: `Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2`.

B1-S2 Screenshot: Shows the output of the command `Switch# sh etherchannel summary`. It displays the status of EtherChannel groups 1 and 2, including flags, number of channel-groups in use, and number of aggregators. A red line highlights the ports for group 2: `Fa0/1(P) Fa0/2(P)`.

B1-S1 Screenshot: Shows the output of the command `Switch#sh etherchannel summary`. It displays the status of EtherChannel groups 1 and 2, including flags, number of channel-groups in use, and number of aggregators. A red line highlights the ports for group 2: `Fa0/1(P) Fa0/2(P)`.

Étape 2 : configuration de la sécurité des ports

Utilisez la stratégie suivante pour activer la sécurité des ports sur les ports d'accès BX-S2 :

- Autoriser une seule adresse MAC
- Forcer la première adresse MAC apprise à s'adapter à la configuration
- Forcer la désactivation du port en cas de violation de la sécurité

B1-S2

Physical Config CLI Attributes

IOS Command Line Interface

```

switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.BCCA.9C3A
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.9747.CE95
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.4339.0C91
!

```

B2-S2

Physical Config CLI Attributes

IOS Command Line Interface

```

!
interface FastEthernet0/6
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 000C.CFD8.751C
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.64D9.7C5C
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0002.1664.72A0
!

```

B3-S2

Physical Config **CLI** Attributes

IOS Command Line Interface

```

!
interface FastEthernet0/6
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
!

```

Étape 3 : vérification des affectations VLAN et de la sécurité des ports

Utilisez les commandes adéquates pour vérifier que les réseaux locaux virtuels (VLAN) d'accès sont correctement affectés et que la stratégie de sécurité des ports a été activée.

B1-S2

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Switch#sh port-security int fa
Switch#sh port-security int fastEthernet 0/6
Port Security                : Enabled
Port Status                  : Secure-up
Violation Mode                : Shutdown
Aging Time                   : 0 mins
Aging Type                   : Absolute
SecureStatic Address Aging   : Disabled
Maximum MAC Addresses        : 1
Total MAC Addresses          : 1
Configured MAC Addresses     : 0
Sticky MAC Addresses         : 1
Last Source Address:Vlan    : 00D0.BCCA.9C3A:10
Security Violation Count     : 0

Switch#sh port-security int fastEthernet 0/11
Port Security                : Enabled
Port Status                  : Secure-up
Violation Mode                : Shutdown
Aging Time                   : 0 mins
Aging Type                   : Absolute
SecureStatic Address Aging   : Disabled
Maximum MAC Addresses        : 1
Total MAC Addresses          : 1
Configured MAC Addresses     : 0
Sticky MAC Addresses         : 1
Last Source Address:Vlan    : 0001.9747.CE95:20
Security Violation Count     : 0

Switch#sh port-security int fastEthernet 0/16
Port Security                : Enabled
Port Status                  : Secure-up
Violation Mode                : Shutdown
Aging Time                   : 0 mins
Aging Type                   : Absolute
SecureStatic Address Aging   : Disabled
Maximum MAC Addresses        : 1

```

Tâche 9 : configuration du protocole STP

Étape 1 : configuration de BX-S1 comme pont racine

Définissez le niveau de priorité sur 4096 pour BX-S1, de manière à ce que ces commutateurs restent en permanence le pont racine de tous les réseaux locaux virtuels.

B1-S1(config)#spanning-tree vlan 1-1001 priority 4096

B2-S1(config)#spanning-tree vlan 1-1001 priority 4096

B3-S1(config)#spanning-tree vlan 1-1001 priority 4096

Étape 2 : configuration de BX-S3 comme pont racine de secours

Définissez le niveau de priorité sur 8192 pour BX-S3, de manière à ce que ces commutateurs restent en permanence le pont racine de secours de tous les réseaux locaux virtuels.

B1-S3(config)#spanning-tree vlan 1-1001 priority 8192

B2-S3(config)#spanning-tree vlan 1-1001 priority 8192

B3-S3(config)#spanning-tree vlan 1-1001 priority 8192

Étape 3 : vérification que BX-S1 est le pont racine

```

Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address    0001.9665.A9E9
            Cost        12
            Port        28 (Port-channel2)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    00D0.BA3D.2C94
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Po1          Altn BLK 12        128.27   Shr
Fa0/5        Desg FWD 19        128.5    P2p
Po2          Root FWD 12        128.28   Shr

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
            Address    0001.9665.A9E9
            Cost        12
            Port        28 (Port-channel2)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
            Address    00D0.BA3D.2C94
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Po1          Altn BLK 12        128.27   Shr
Fa0/5        Desg FWD 19        128.5    P2p
Po2          Root FWD 12        128.28   Shr
  
```

```

Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address    0010.1135.2774
            Cost        12
            Port        27 (Port-channel1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0090.2115.B847
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/5        Desg FWD 19      128.5    P2p
Po1          Desg FWD 12      128.27   Shr
Po2          Desg FWD 12      128.28   Shr

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
            Address    0010.1135.2774
            Cost        12
            Port        27 (Port-channel1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
            Address    0090.2115.B847
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/5        Desg FWD 19      128.5    P2p
Po1          Desg FWD 12      128.27   Shr
Po2          Desg FWD 12      128.28   Shr
  
```

```

Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address    0002.17B7.6BC1
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0002.17B7.6BC1
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/5        Desg FWD 19      128.5    P2p
Po1          Desg FWD 12      128.27   Shr
Po2          Desg FWD 12      128.28   Shr

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
            Address    0002.17B7.6BC1
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
            Address    0002.17B7.6BC1
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

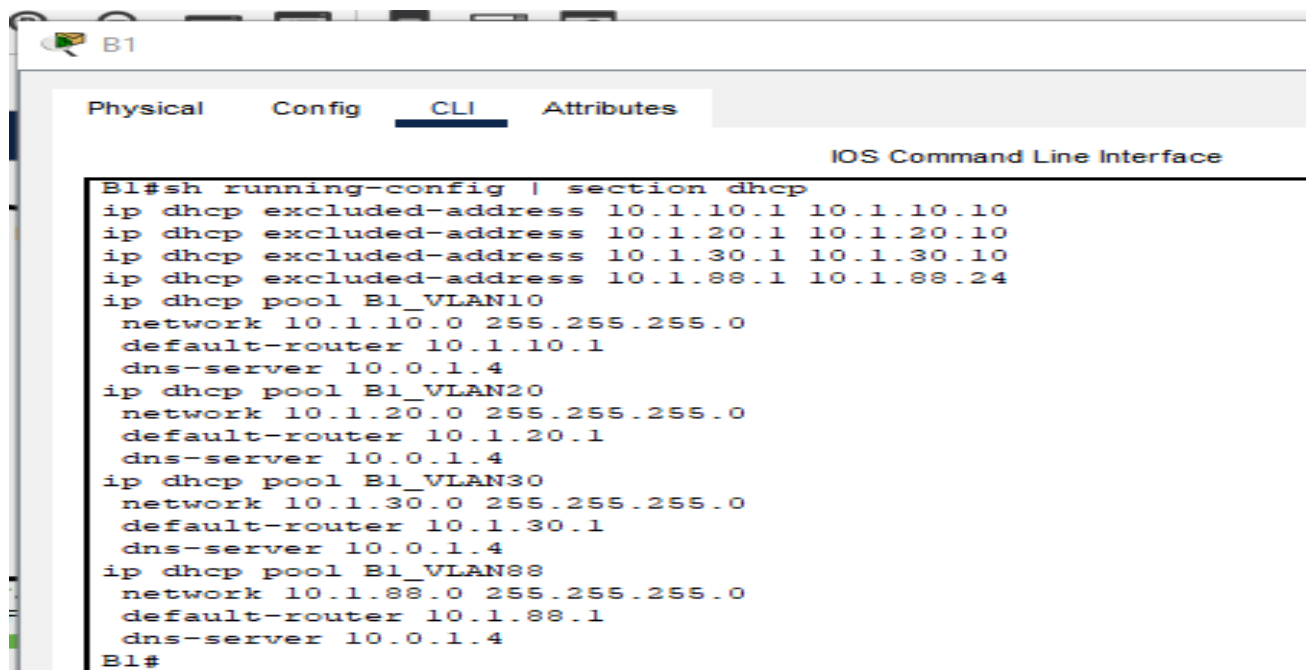
Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/5        Desg FWD 19      128.5    P2p
Po1          Desg FWD 12      128.27   Shr
Po2          Desg FWD 12      128.28   Shr
  
```


Tâche 10 : configuration du service DHCP

Étape 1 : configuration de pools DHCP pour chaque réseau local virtuel

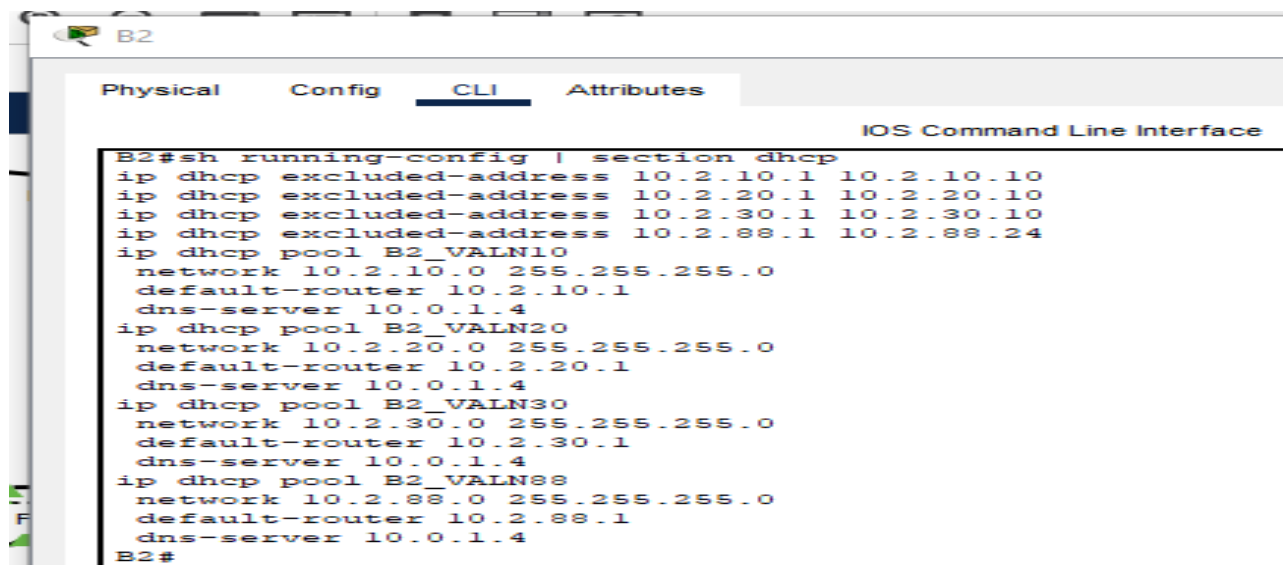
Sur les routeurs Branch, configurez des pools DHCP pour chaque réseau local virtuel, en respectant les consignes suivantes :

- Excluez les dix premières adresses IP de chaque pool pour les réserver aux réseaux locaux.
 - Excluez les 24 premières adresses IP de chaque pool pour les réserver aux réseaux locaux sans fil.
 - Le nom du pool est BX_VLAN##, où X est le numéro du routeur et ## est le numéro du réseau local virtuel.
- Dans le cadre de votre configuration DHCP, incluez le serveur DNS connecté à la batterie de serveurs HQ.



```

B1#sh running-config | section dhcp
ip dhcp excluded-address 10.1.10.1 10.1.10.10
ip dhcp excluded-address 10.1.20.1 10.1.20.10
ip dhcp excluded-address 10.1.30.1 10.1.30.10
ip dhcp excluded-address 10.1.88.1 10.1.88.24
ip dhcp pool B1_VLAN10
  network 10.1.10.0 255.255.255.0
  default-router 10.1.10.1
  dns-server 10.0.1.4
ip dhcp pool B1_VLAN20
  network 10.1.20.0 255.255.255.0
  default-router 10.1.20.1
  dns-server 10.0.1.4
ip dhcp pool B1_VLAN30
  network 10.1.30.0 255.255.255.0
  default-router 10.1.30.1
  dns-server 10.0.1.4
ip dhcp pool B1_VLAN88
  network 10.1.88.0 255.255.255.0
  default-router 10.1.88.1
  dns-server 10.0.1.4
B1#
  
```

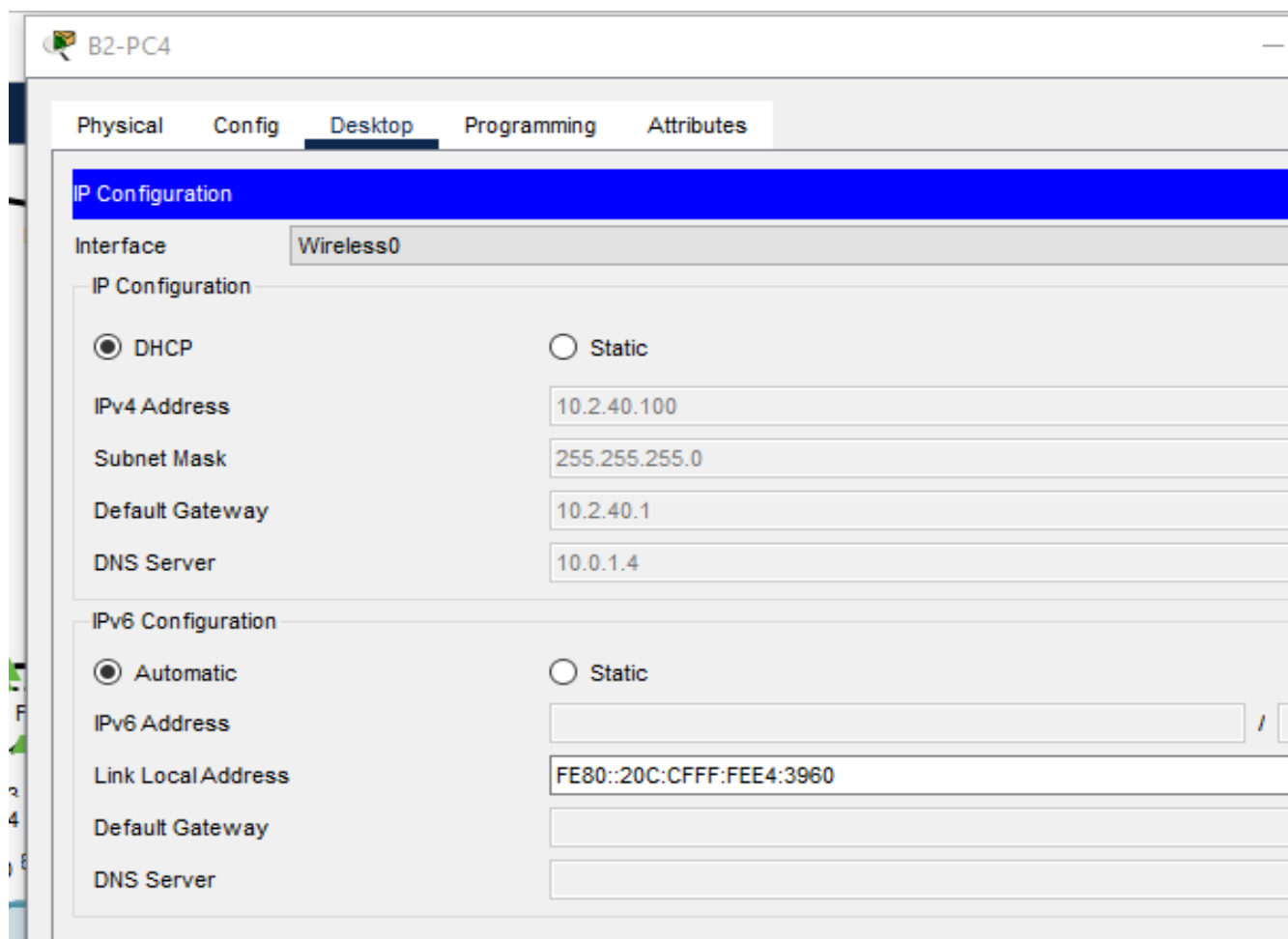
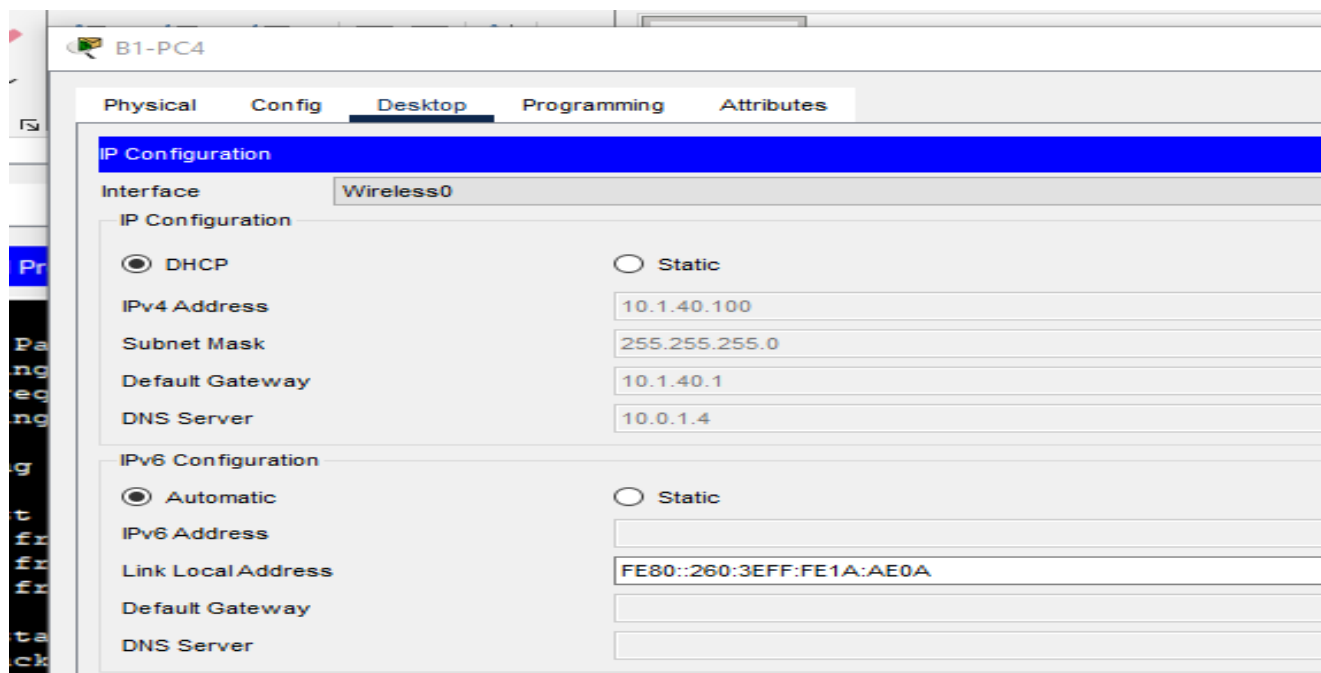


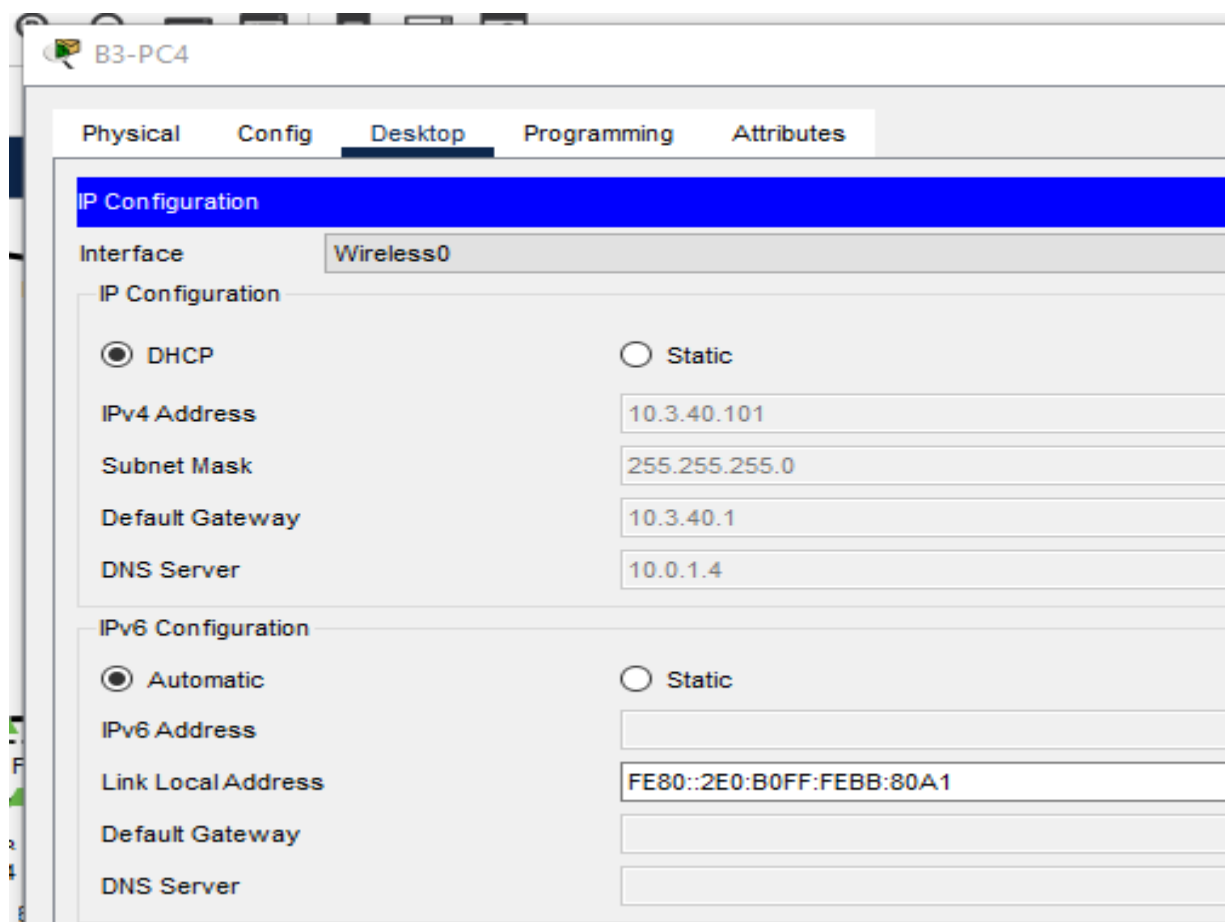
```

B2#sh running-config | section dhcp
ip dhcp excluded-address 10.2.10.1 10.2.10.10
ip dhcp excluded-address 10.2.20.1 10.2.20.10
ip dhcp excluded-address 10.2.30.1 10.2.30.10
ip dhcp excluded-address 10.2.88.1 10.2.88.24
ip dhcp pool B2_VALN10
  network 10.2.10.0 255.255.255.0
  default-router 10.2.10.1
  dns-server 10.0.1.4
ip dhcp pool B2_VALN20
  network 10.2.20.0 255.255.255.0
  default-router 10.2.20.1
  dns-server 10.0.1.4
ip dhcp pool B2_VALN30
  network 10.2.30.0 255.255.255.0
  default-router 10.2.30.1
  dns-server 10.0.1.4
ip dhcp pool B2_VALN88
  network 10.2.88.0 255.255.255.0
  default-router 10.2.88.1
  dns-server 10.0.1.4
B2#
  
```

Étape 2 : activation de DHCP sur les PC

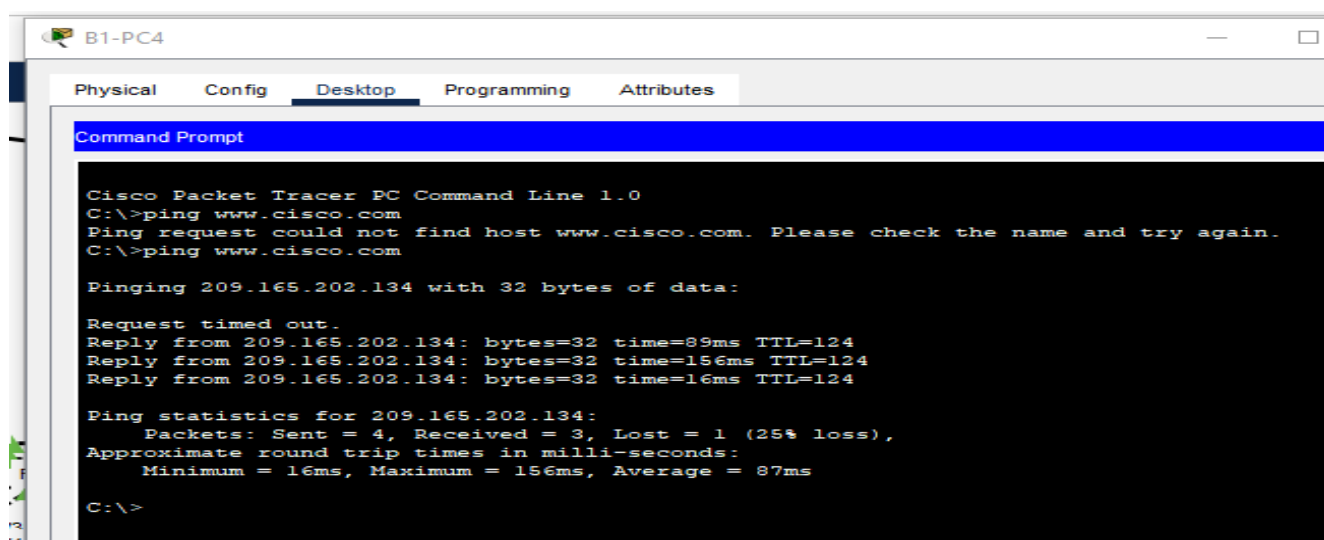
Actuellement, les ordinateurs sont configurés pour utiliser des adresses IP statiques. Modifiez cette configuration de sorte qu'ils utilisent le service DHCP

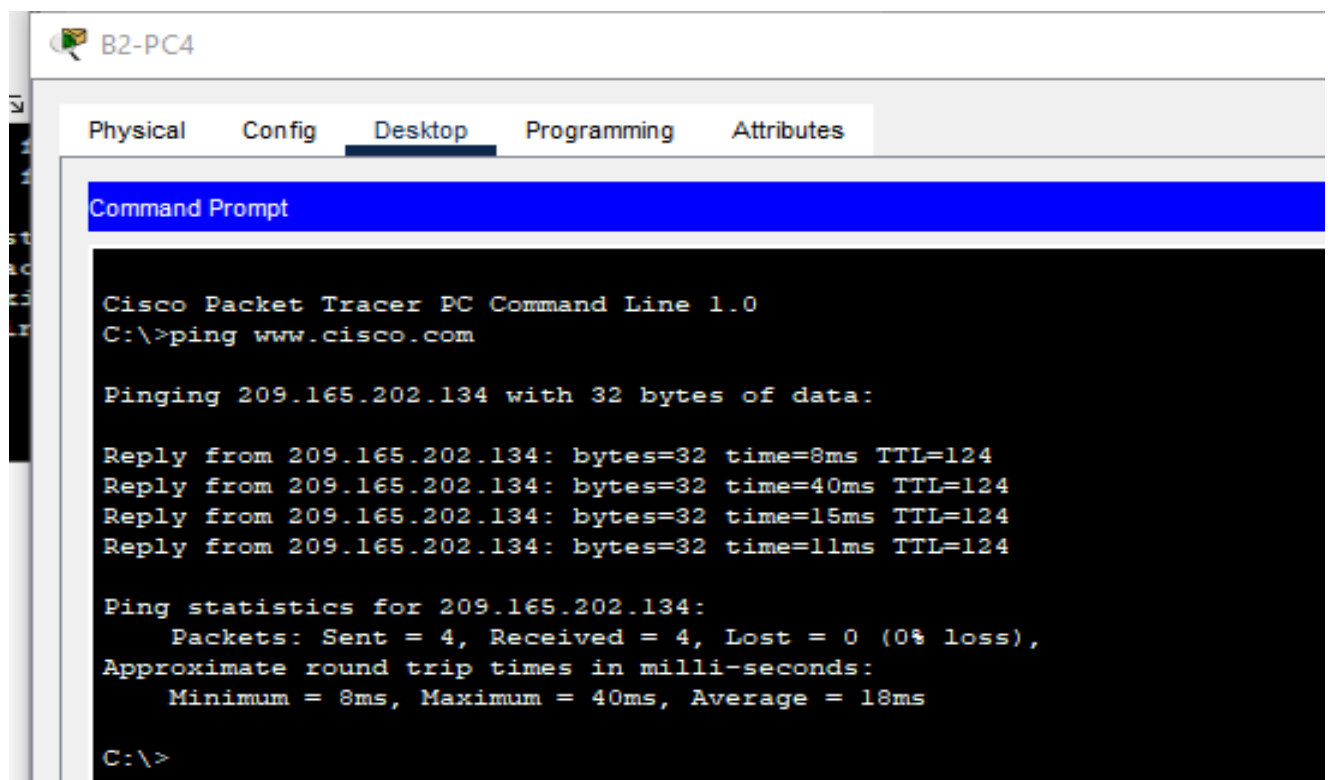




Étape 4 : vérification de la connectivité

Tous les PC physiquement connectés au réseau doivent pouvoir envoyer des requêtes ping vers le serveur Web www.cisco.com.





B2-PC4

Physical Config **Desktop** Programming Attributes

Command Prompt

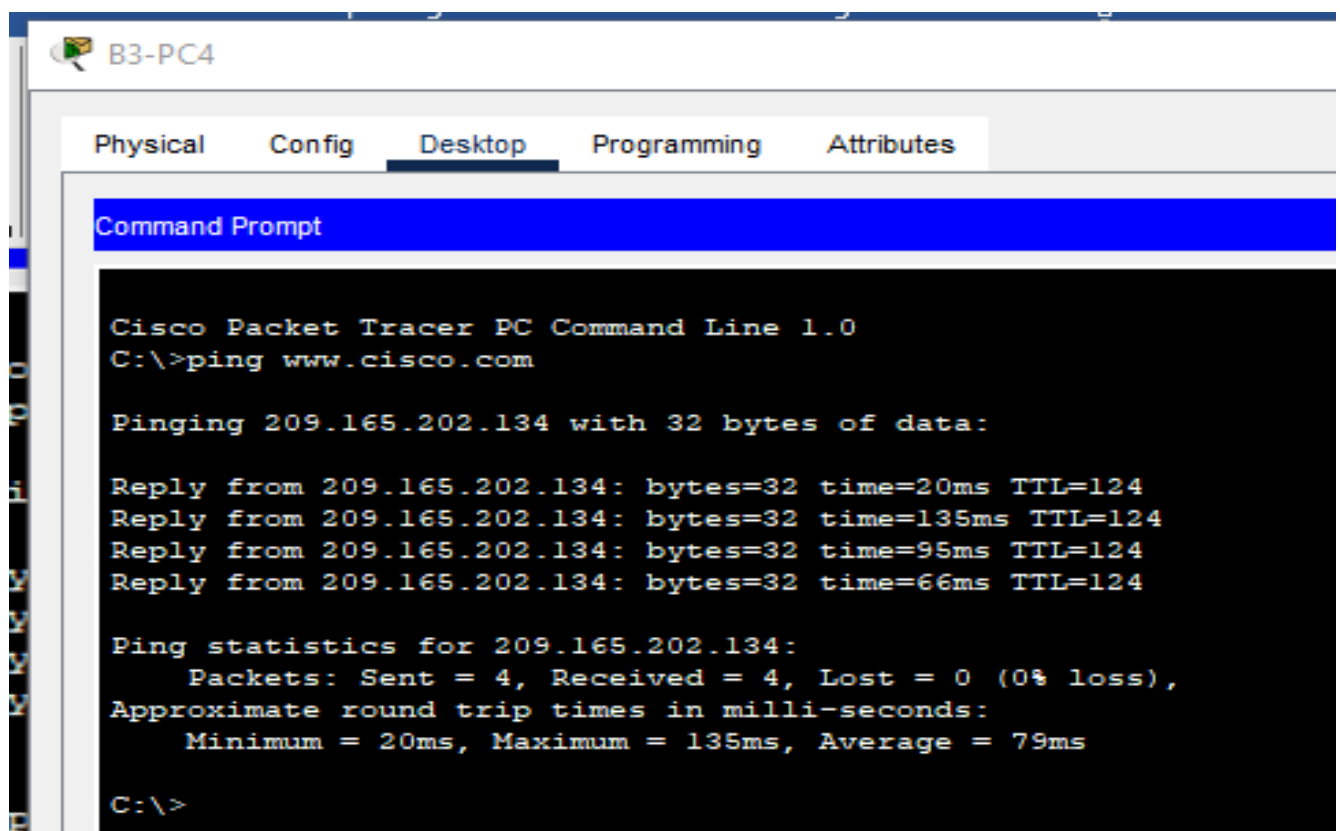
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=8ms TTL=124
Reply from 209.165.202.134: bytes=32 time=40ms TTL=124
Reply from 209.165.202.134: bytes=32 time=15ms TTL=124
Reply from 209.165.202.134: bytes=32 time=11ms TTL=124

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 40ms, Average = 18ms

C:\>
```



B3-PC4

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=20ms TTL=124
Reply from 209.165.202.134: bytes=32 time=135ms TTL=124
Reply from 209.165.202.134: bytes=32 time=95ms TTL=124
Reply from 209.165.202.134: bytes=32 time=66ms TTL=124

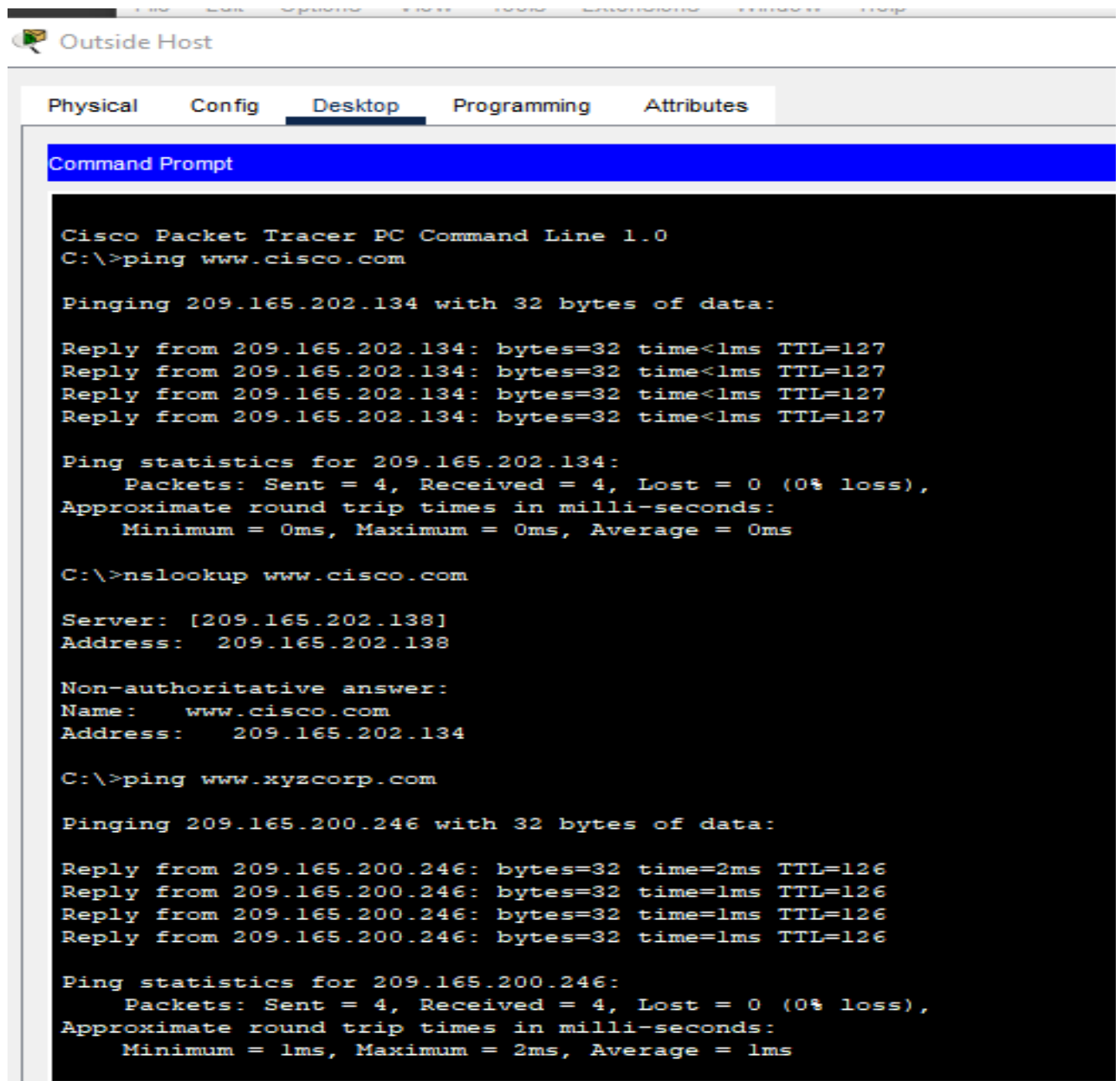
Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 135ms, Average = 79ms

C:\>
```

Tâche 11 : configuration d'une liste de contrôle d'accès pare-feu

Étape 1 : vérification de la connectivité à partir d'Outside Host

Le PC Outside Host doit être capable d'envoyer des requêtes ping au serveur à l'adresse www.xyzcorp.com.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>nslookup www.cisco.com

Server: [209.165.202.138]
Address: 209.165.202.138

Non-authoritative answer:
Name: www.cisco.com
Address: 209.165.202.134

C:\>ping www.xyzcorp.com

Pinging 209.165.200.246 with 32 bytes of data:

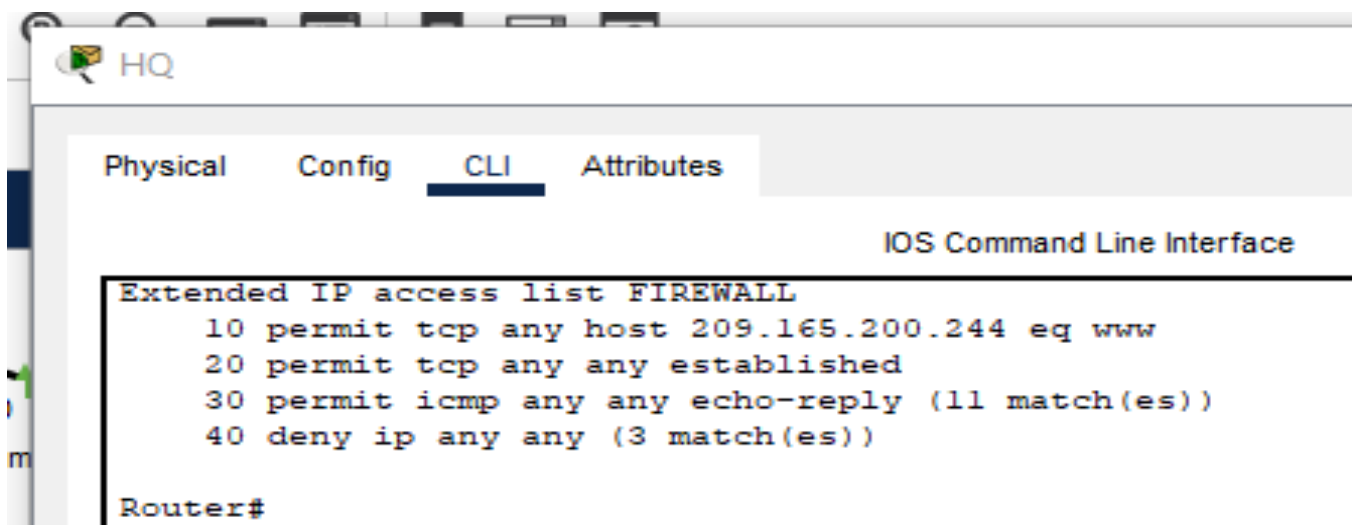
Reply from 209.165.200.246: bytes=32 time=2ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.200.246:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Étape 2 : mise en œuvre d'une liste de contrôle d'accès pare-feu rudimentaire

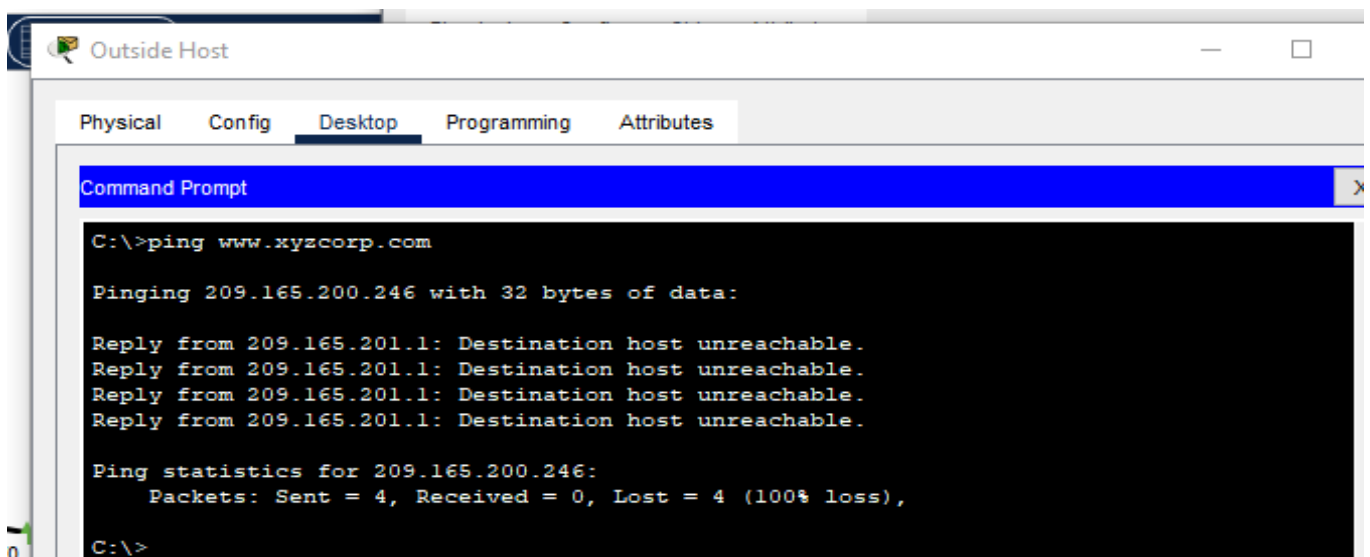
ISP permettant la connexion à Internet, configurez une liste de contrôle d'accès nommée, appelée FIREWALL, en ajoutant les instructions dans l'ordre suivant :

1. Autorisez les requêtes HTTP entrantes vers le serveur www.xyzcorp.com.
2. Autorisez uniquement les sessions TCP établies provenant d'ISP et des sources situées au-delà d'ISP.
3. Autorisez uniquement les réponses ping entrantes provenant d'ISP et des sources situées au-delà d'ISP.
4. Bloquez de manière explicite tous les autres accès entrants provenant d'ISP et des sources situées au-delà d'ISP



Étape 3 : vérification de la connectivité à partir d'Outside Host

Le PC Outside Host ne doit pas être en mesure d'envoyer des requêtes ping vers le serveur à l'adresse www.xyzcorp.com. En revanche, le PC Outside Host doit pouvoir envoyer des requêtes d'affichage de pages Web.



Tâche 12 : configuration de la connectivité sans fil

Étape 1 : vérification de la configuration DHCP

Chaque routeur BX-WRS doit déjà disposer d'un adressage IP obtenu auprès du service DHCP du routeur BX pour le réseau local virtuel VLAN 88.

B1-WRS

Physical **Config** GUI Attributes

GLOBAL

- Settings
- Algorithm Settings

INTERFACE

- Internet**
- LAN
- Wireless

Internet Settings

IP Configuration

- ☒ DHCP
- ☐ Static
- ☐ PPPoE

UserName

Password

IPv4 Address: 10.1.88.25

Subnet Mask: 255.255.255.0

Default Gateway: 10.1.88.1

DNS Server: 10.0.1.4

B2-WRS

Physical **Config** GUI Attributes

GLOBAL

- Settings
- Algorithm Settings

INTERFACE

- Internet**
- LAN
- Wireless

Internet Settings

IP Configuration

- ☒ DHCP
- ☐ Static
- ☐ PPPoE

UserName

Password

IPv4 Address: 10.2.88.25

Subnet Mask: 255.255.255.0

Default Gateway: 10.2.88.1

DNS Server: 10.0.1.4

B3-WRS

Physical **Config** GUI Attributes

GLOBAL

- Settings
- Algorithm Settings

INTERFACE

- Internet**
- LAN
- Wireless

Internet Settings

IP Configuration

- ☒ DHCP
- ☐ Static
- ☐ PPPoE

UserName

Password

IPv4 Address: 10.3.88.25

Subnet Mask: 255.255.255.0

Default Gateway: 10.3.88.1

DNS Server: 10.0.1.4

Étape 2 : configuration des paramètres de configuration réseau et de réseau local

Sous l'onglet GUI, le champ « Router IP » de la page Status doit correspondre à la première adresse IP du sous-réseau 10.X.40.0 /24. Conservez les valeurs par défaut des autres paramètres.

The screenshot shows the configuration interface for B1-WRS. The 'GUI' tab is selected. The 'Internet Setup' section is active, showing 'Automatic Configuration - DHCP' as the Internet Connection type. The 'Network Setup' section shows the Router IP configuration. The IP Address is set to 10.1.40.1, and the Subnet Mask is 255.255.255.0. The DHCP Server is enabled, and the Start IP Address is 10.1.40.100. The Maximum number of Users is 50.

The screenshot shows the configuration interface for B2-WRS. The 'GUI' tab is selected. The 'Internet Setup' section is active, showing 'Automatic Configuration - DHCP' as the Internet Connection type. The 'Network Setup' section shows the Router IP configuration. The IP Address is set to 10.2.40.1, and the Subnet Mask is 255.255.255.0. The DHCP Server is enabled, and the Start IP Address is 10.2.40.100. The Maximum number of Users is 50.

The screenshot shows the configuration interface for a B3-WRS router. The 'Setup' tab is selected, and the 'Internet Setup' section is active. The 'Internet Connection type' is set to 'Automatic Configuration - DHCP'. Under 'Optional Settings', the 'Host Name' and 'Domain Name' fields are empty, and the 'MTU' is set to 1500. In the 'Network Setup' section, the 'Router IP' is configured as 10.3.40.1 with a 'Subnet Mask' of 255.255.255.0. The 'DHCP Server' is enabled, and the 'Start IP Address' is 10.3.40.100, with a 'Maximum number of Users' set to 50. A 'DHCP Reservation' button is visible on the right.

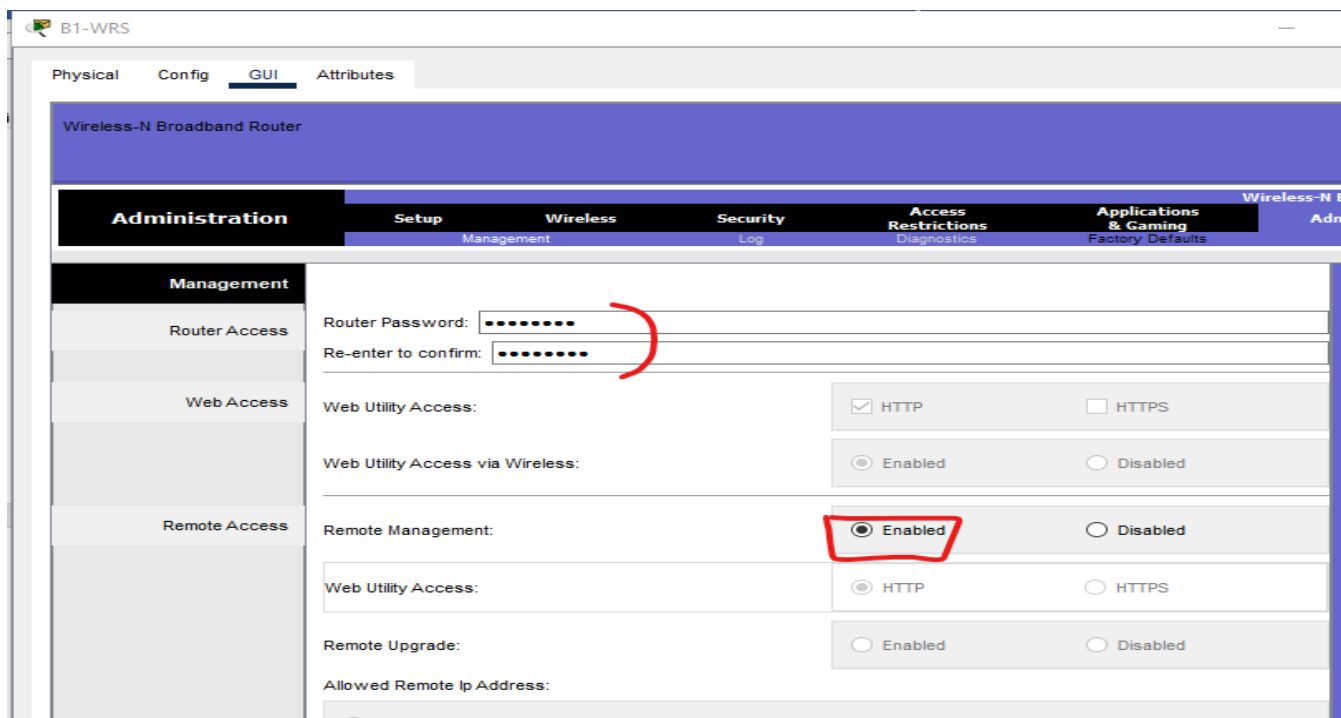
Étape 3 : configuration des paramètres de configuration de réseau sans fil

Les SSID des routeurs sont BX-WRS_LAN, où X correspond au numéro du routeur Branch.
La clé WEP est 12345ABCDE.

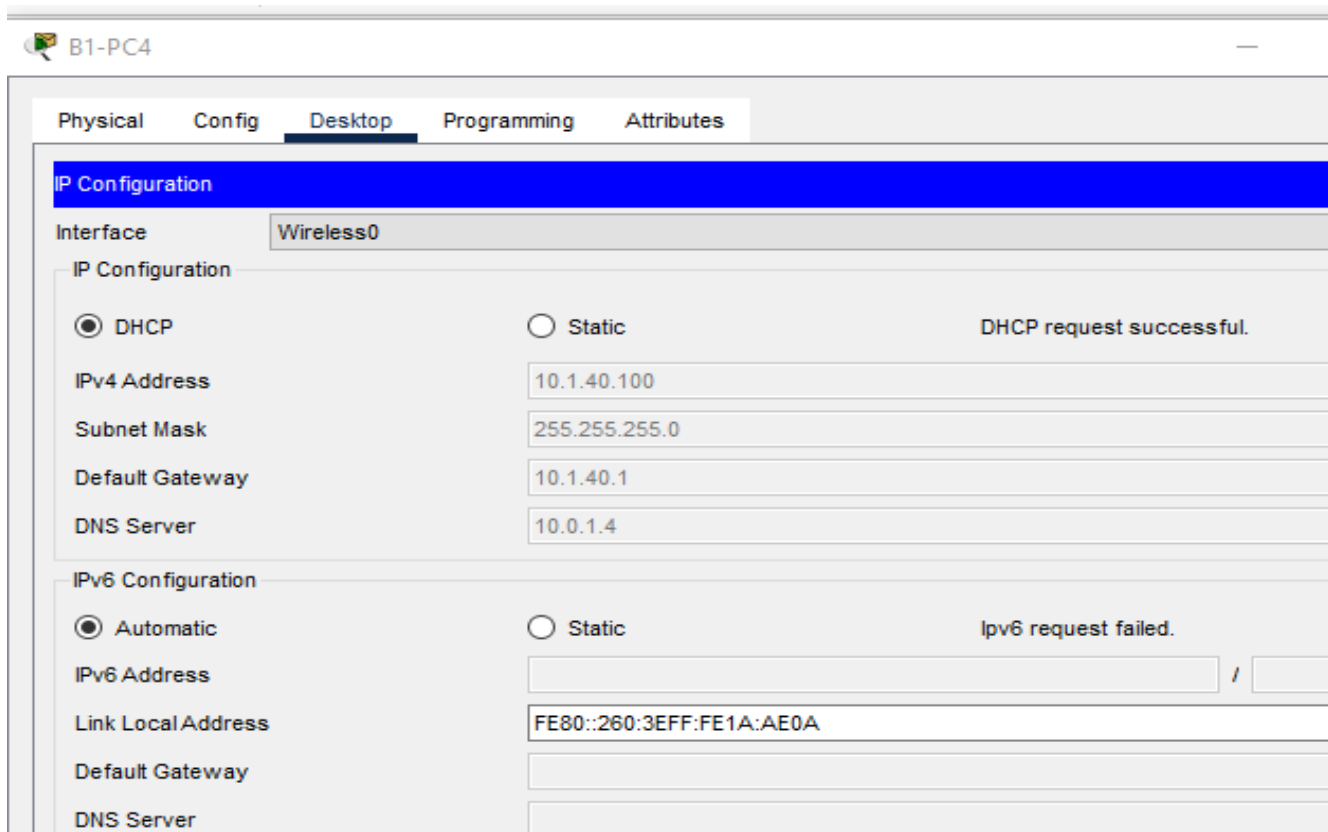
The screenshot shows the configuration interface for a B1-WRS router. The 'Wireless' tab is selected, and the 'Wireless Security' section is active. The 'Security Mode' is set to 'WEP'. The 'Encryption' is set to '40/64-Bits (10 Hex digits)'. A 'Passphrase' field is present with a 'Generate' button. Below, four 'Key' fields are shown, with 'Key1' containing the value '12345ABCDE' and the other keys being empty.

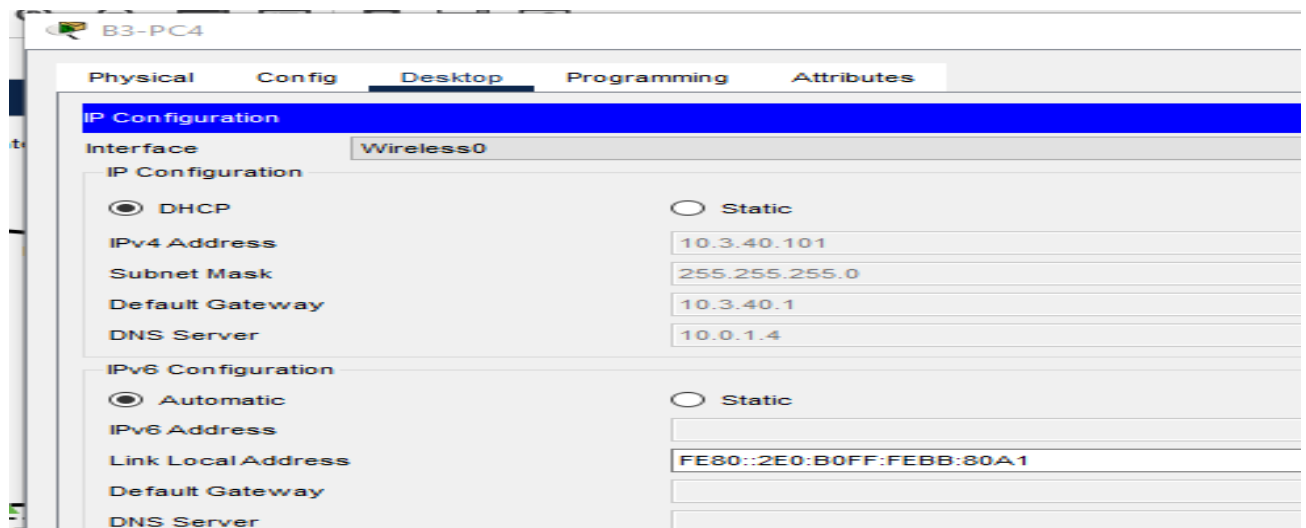
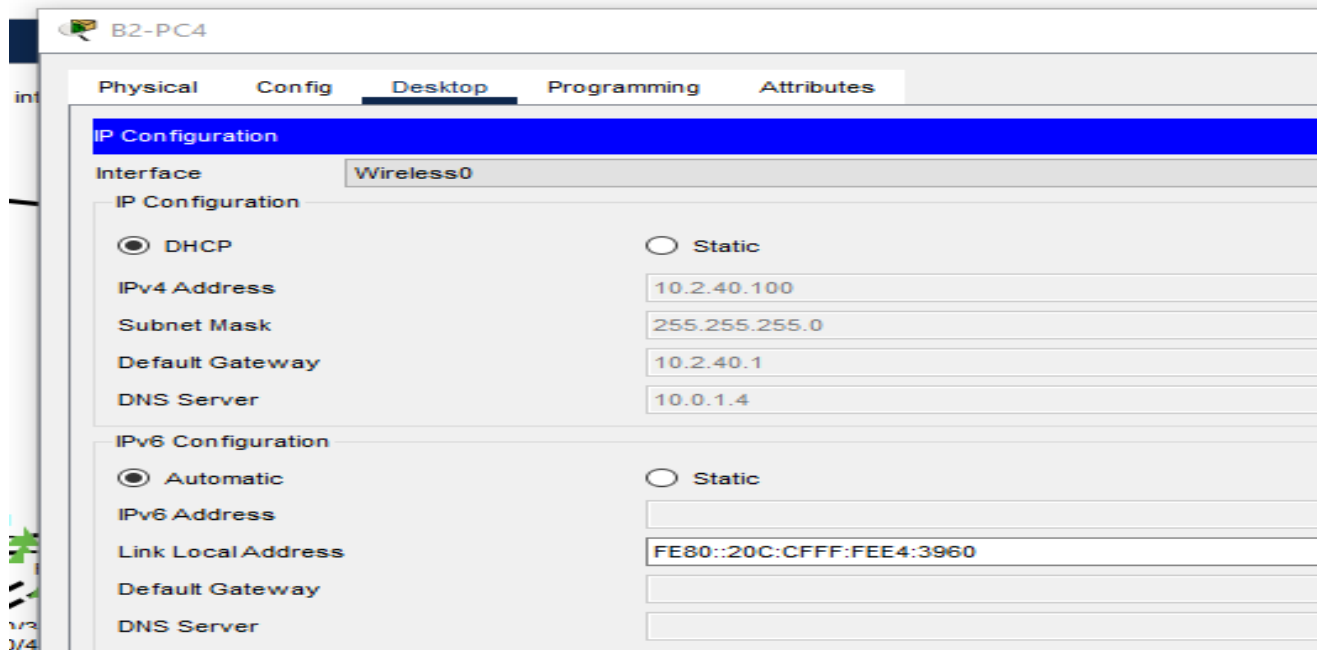
Étape 4 : configurez l'accès distant sur les routeurs sans fil.

Définissez cisco123 comme mot de passe d'administration et activez l'administration à distance.



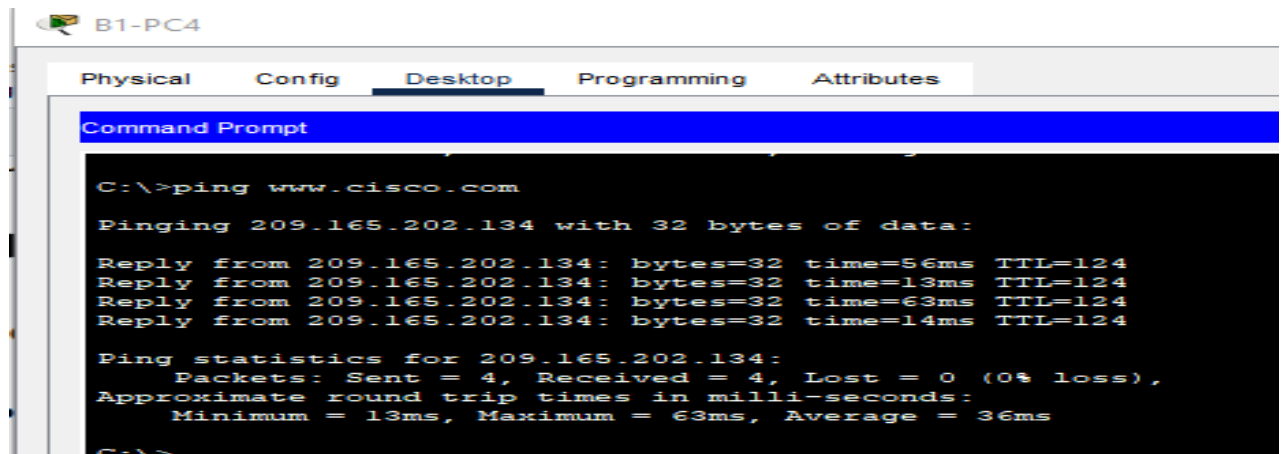
Étape 5 : configuration des PC BX-PC4 pour qu'ils accèdent au réseau sans fil par le biais du service DHCP

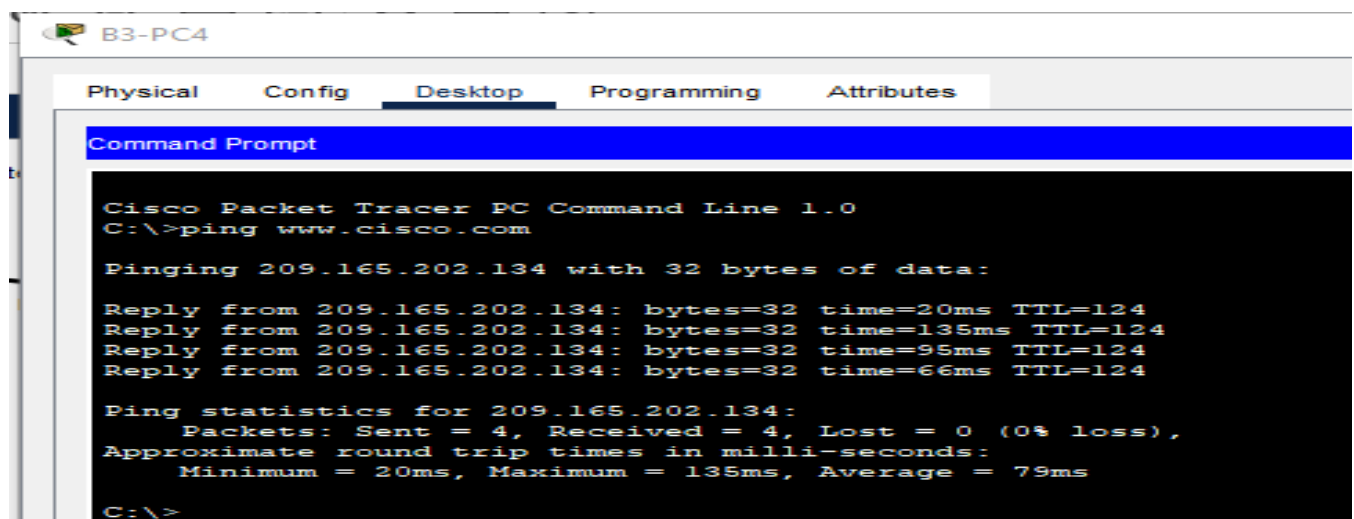
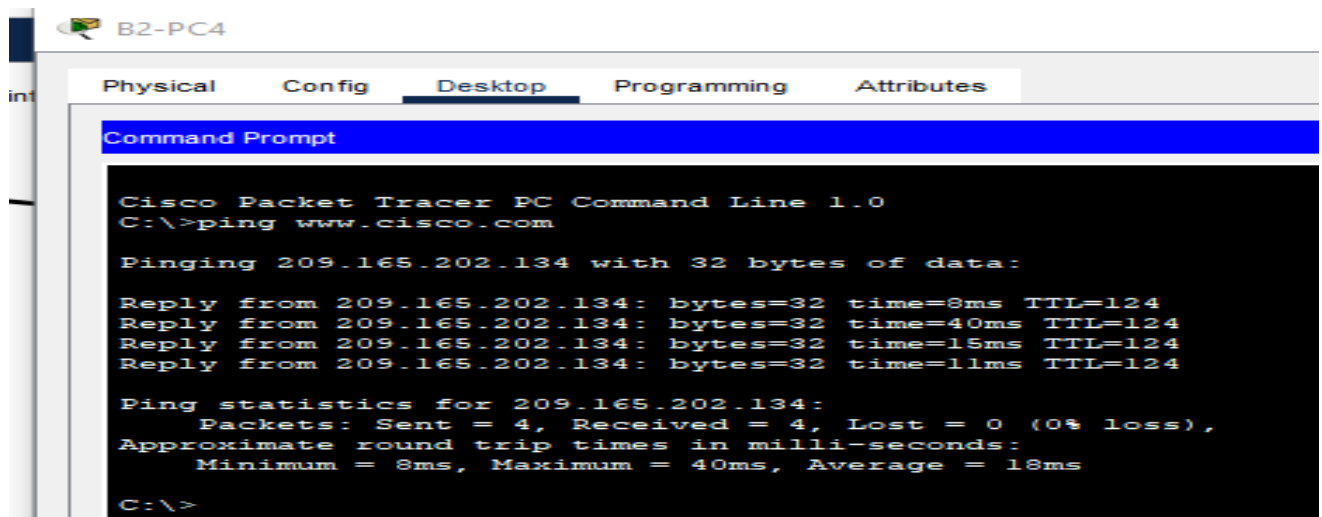




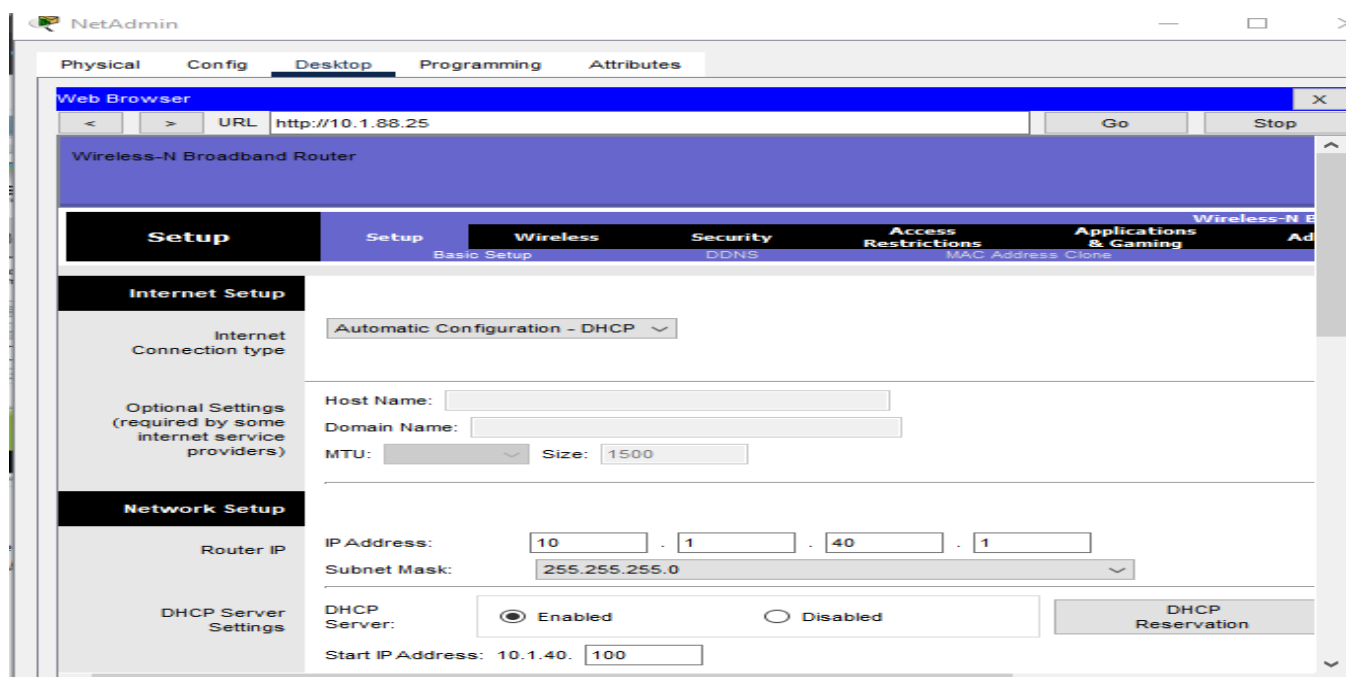
Etape 6 : vérification de la connectivité et du fonctionnement de l'administration à distance

Tous les PC sans fil doivent pouvoir accéder au serveur Web www.cisco.com.





Vérifiez le fonctionnement de l'administration à distance en accédant au routeur sans fil par le biais du navigateur Web.



NetAdmin

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://10.2.88.25 Go Stop

Wireless-N Broadband Router

Setup Setup Wireless Security Access Restrictions Applications & Gaming Ad

Basic Setup DDNS MAC Address Clone

Internet Setup

Internet Connection type Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP IP Address: 10 . 2 . 40 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 10.2.40. 100

NetAdmin

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://10.3.88.25 Go Stop

Wireless-N Broadband Router

Setup Setup Wireless Security Access Restrictions Applications & Gaming Ad

Basic Setup DDNS MAC Address Clone

Internet Setup

Internet Connection type Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name: Domain Name: MTU: Size: 1500

Network Setup

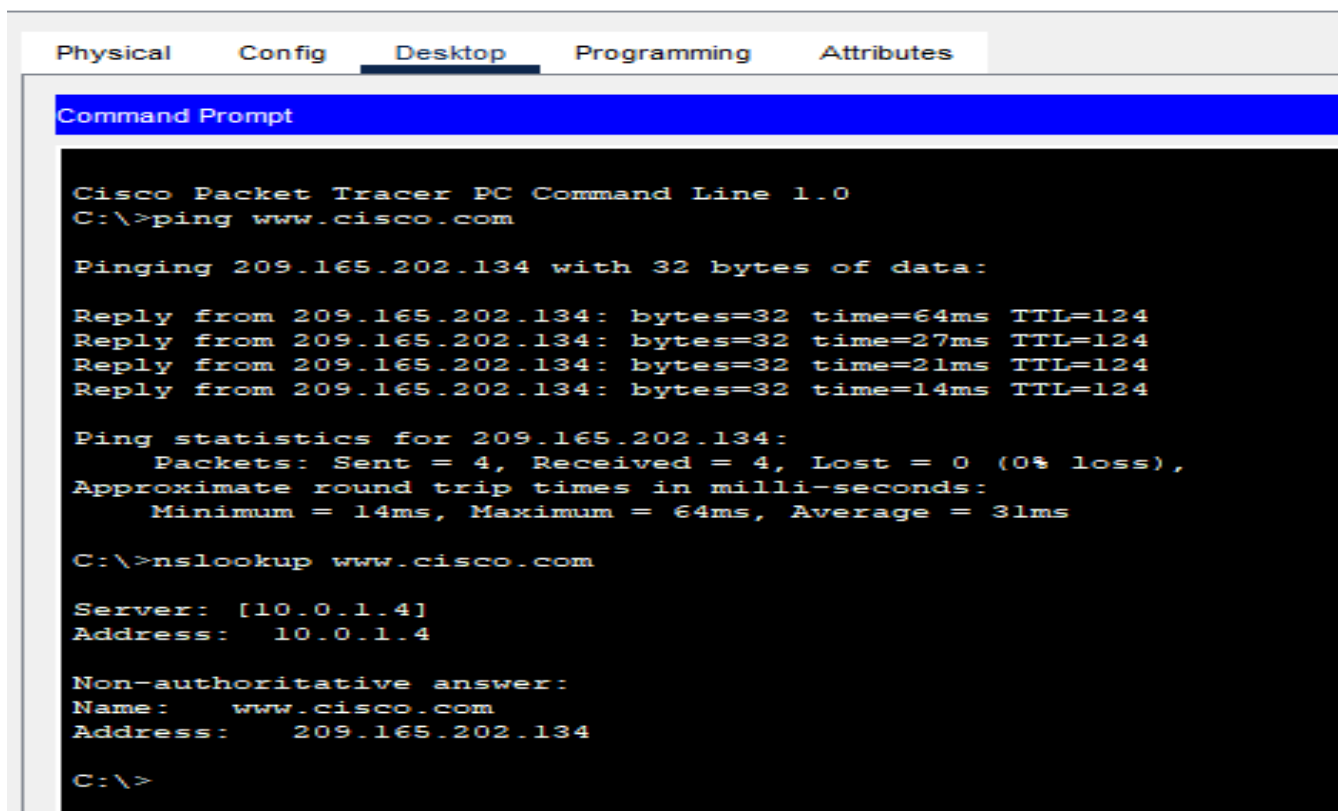
Router IP IP Address: 10 . 3 . 40 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 10.3.40. 100

☐ Top

B2-PC4



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=64ms TTL=124
Reply from 209.165.202.134: bytes=32 time=27ms TTL=124
Reply from 209.165.202.134: bytes=32 time=21ms TTL=124
Reply from 209.165.202.134: bytes=32 time=14ms TTL=124

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 64ms, Average = 31ms

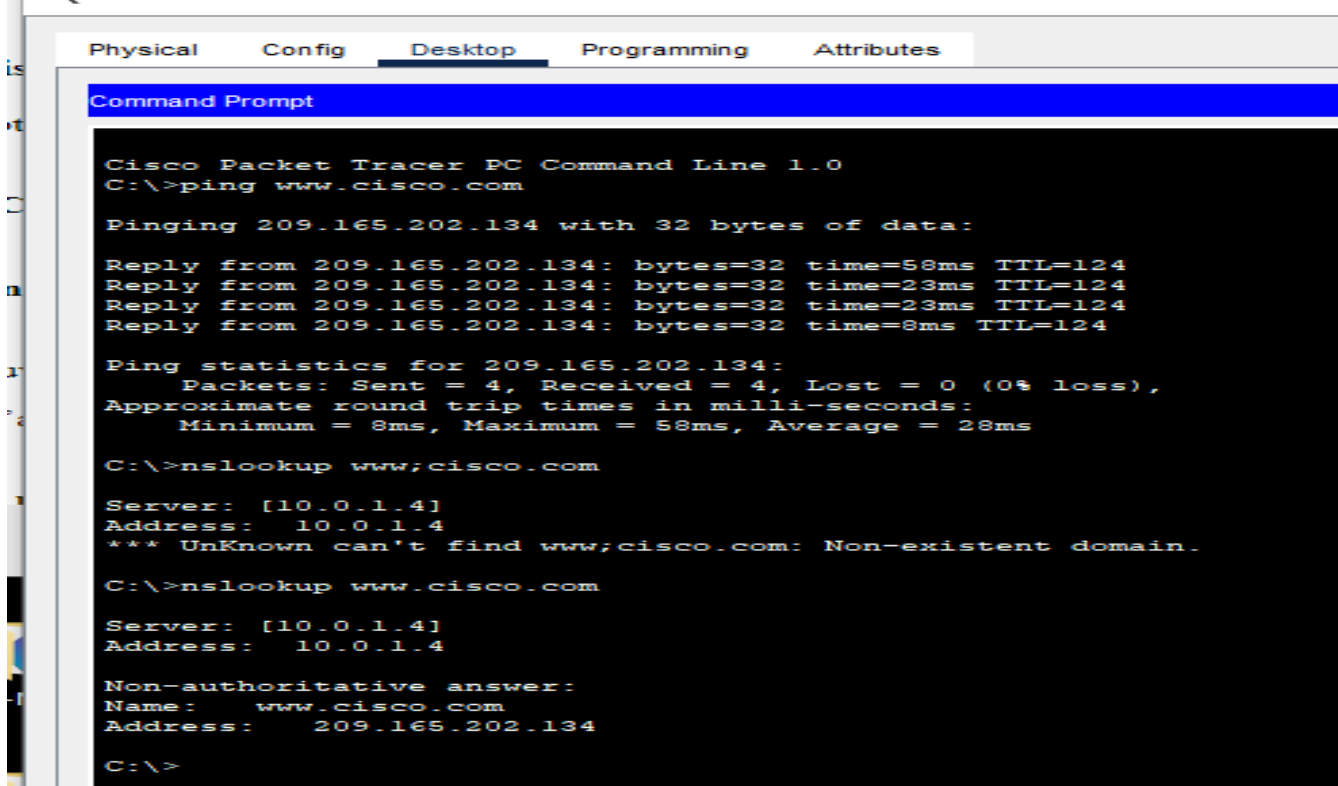
C:\>nslookup www.cisco.com

Server: [10.0.1.4]
Address: 10.0.1.4

Non-authoritative answer:
Name: www.cisco.com
Address: 209.165.202.134

C:\>
```

B3-PC4



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time=58ms TTL=124
Reply from 209.165.202.134: bytes=32 time=23ms TTL=124
Reply from 209.165.202.134: bytes=32 time=23ms TTL=124
Reply from 209.165.202.134: bytes=32 time=8ms TTL=124

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 58ms, Average = 28ms

C:\>nslookup www;cisco.com

Server: [10.0.1.4]
Address: 10.0.1.4
*** UnKnown can't find www;cisco.com: Non-existent domain.

C:\>nslookup www.cisco.com

Server: [10.0.1.4]
Address: 10.0.1.4

Non-authoritative answer:
Name: www.cisco.com
Address: 209.165.202.134

C:\>
```